

# **McIntire School Expansion: Project Execution Plan and Risk Assessment**

A Technical Report submitted to the Department of Civil and Environmental Engineering

Presented to the Faculty of the School of Engineering and Applied Science  
University of Virginia • Charlottesville, Virginia

In Partial Fulfillment of the Requirements for the Degree  
Bachelor of Science, School of Engineering

**Saul Joseph Malca Hasky**

Spring, 2026

Technical Project Team Members

Stephanie Castro

Timothy Cheng

Griffin Harrison

Aden Melton

On my honor as a University Student, I have neither given nor received unauthorized aid on this assignment as defined by the Honor Guidelines for Thesis-Related Assignments

Rob Tieman, Department of Civil and Environmental Engineering

Matt O'Malley, Department of Civil and Environmental Engineering

## Table of Contents

|                                                                          |           |
|--------------------------------------------------------------------------|-----------|
| <b>INTRODUCTION.....</b>                                                 | <b>3</b>  |
| Design Problem Statement.....                                            | 3         |
| Design Objectives .....                                                  | 3         |
| Background.....                                                          | 4         |
| Evolving Project Complexity and Constraints .....                        | 5         |
| <b>DESIGN .....</b>                                                      | <b>9</b>  |
| Support of Excavation Analysis .....                                     | 9         |
| Underpinning Design Process.....                                         | 11        |
| Tunnel Design Considerations and Codes .....                             | 13        |
| Tunnel Sequencing Planning .....                                         | 15        |
| Site Logistics Plan .....                                                | 20        |
| Cast-in-Place Concrete Self-Perform .....                                | 25        |
| Risk Register .....                                                      | 31        |
| <b>CONCLUSION &amp; DISCUSSION.....</b>                                  | <b>34</b> |
| <b>APPENDICES .....</b>                                                  | <b>35</b> |
| Appendix A - Detailed Schedule .....                                     | 35        |
| Appendix B - Design Evolution .....                                      | 38        |
| Appendix C - Engineering Standards Compliance.....                       | 39        |
| Appendix D - Supporting Technical Deliverables .....                     | 41        |
| Appendix E - Clash Detection Details (New Utilities) .....               | 42        |
| Appendix F - Clash Detection Details (Existing Utilities).....           | 46        |
| Appendix G - Existing Utilitiy/Tunnel Sequencing Supporting Graphics.... | 49        |
| Appendix H - Underpinning Support Design Recommendations .....           | 50        |
| Appendix I - Supporting Materials.....                                   | 52        |
| Appendix J - Peer Review .....                                           | 53        |

# INTRODUCTION

## Design Problem Statement

The McIntire Expansion Project presents a complex construction challenge with high risks to cost, schedule, and quality. The project combines the renovation of Cobb Hall, a century-old structure requiring underpinning and hazardous material abatement, with the construction of the new Shumway Hall, a five-story building featuring a curved structure, a below-grade atrium, and a tunnel connection to Rouss-Robertson. These elements create major constructability concerns. First, the site itself adds significant logistical issues. Construction must occur on a constrained, active campus with limited laydown space, ongoing academic and nearby medical operations, and with multiple participants. Existing utilities, including high-voltage ductbanks and manholes, cross the footprint of the new facility and must be addressed before construction can proceed. On top of these challenges, the project must deliver a design that balances traditional and contemporary architecture to align with UVA's standards, while meeting the fixed deadline of opening Cobb and Shumway Halls for the Fall 2025 semester. The purpose of our project is to act as the contractor/CM at Risk to develop a comprehensive execution plan that addresses these risks: managing schedule pressures, controlling costs amid escalation and supply chain uncertainty, and ensuring quality and safety throughout construction.

## Design Objectives

The capstone team will act as the general contractor/CM at Risk to develop a comprehensive construction plan for the expansion of the McIntire School of Commerce. The team will review all design documents and prepare a detailed project execution plan that includes scheduling, structural design, and coordination of construction activities. Key tasks include evaluating underpinning options for Cobb Hall, planning the construction of Shumway Hall and the tunnel connection to Rouss-Robertson, developing site logistics for the constrained campus environment, and estimating the cost of the concrete scope.

To manage these tasks, the team has developed a coordinated schedule that sequences all deliverables and scope items according to design dependencies, site constraints, and the logical order of construction activities, as shown in Appendix A. All work will follow the engineering standards and technical requirements outlined in Appendix C. The team will also identify major risks related to schedule, cost, safety, and utility conflicts and propose mitigation strategies. Specific items within the scope include:

- **Perform a cost and schedule analysis of multiple underpinning and support options** for the existing foundations of Cobb Hall. Prepare a preliminary structural design of the recommended support option, including drawings and calculations.
- **Analyze the proposed tunnel connection** to Robertson Hall. Design a detailed, coordinated construction and sequencing plan for the tunnel. Develop a solution for the tunnel's configuration and construction methodology that avoids conflicts with existing utilities. Submit a coordinated 3D model showing all existing and proposed conditions, including temporary support. Prepare a sequence-of-work plan describing the proposed construction phasing for the tunnel.
- **Create a full site logistics plan** for each phase of construction, addressing safety, traffic, phasing, and access constraints for the project and surrounding buildings.
- **Prepare a detailed estimate for self-performing the cast-in-place concrete scope**, including a quantity survey, labor productivity analysis, and a full cost projection.
- **Prepare a project risk register** identifying major risks (not limited to those affecting the general contractor). Rate each risk by likelihood and severity, propose mitigation strategies, and present the team's approach for managing each risk.

## Background

The McIntire Expansion Project aligns with UVA's 2030 Plan, which emphasizes strengthening academic and physical foundations, fostering collaboration, and enhancing the student experience. Cobb Hall, built in 1917, no longer meets the university's growing needs for greater enrollment and expansion of its modern commerce programs. Renovating Cobb Hall while constructing Shumway Hall provides updated classrooms, collaborative spaces, and improved connectivity, supporting UVA's goals for academic excellence and flexible learning

environments. The project balances preserving historic architecture with adding contemporary, functional facilities to meet the demands of the McIntire School now and into the future.

## **Evolving Project Complexity and Constraints**

The initial Complexity Statement developed in Fall 2025 framed the project in broad terms: a campus expansion and renovation shaped by limited site space, phased construction, and the need to coordinate new work with existing facilities. Major scope elements - underpinning, tunnel construction, site logistics, cost estimating, and risk - were identified early, but they were



largely viewed as separate challenges. Although the project was recognized as complex, the ways these components interacted and collectively influenced constructability were not yet fully understood.

As the work advanced, the team developed a more integrated view of complexity, recognizing how design decisions directly shaped construction sequencing, logistics, and overall feasibility. The choice of a secant pile wall system, for example, was not simply a structural solution; it emerged from a combination of site constraints, groundwater behavior, and the need to protect Cobb Hall during excavation. Likewise, the tunnel design required careful coordination with existing utilities, prompting adjustments to alignment and sequencing to avoid conflicts and maintain constructability.

Site logistics quickly became one of the most influential drivers of the design. The constrained footprint, heavy pedestrian activity, and proximity to essential campus routes demanded detailed planning for access, staging, and material delivery. This made clear that constructability considerations were just as critical as technical design requirements. At the same time, developing the concrete estimate and risk register highlighted how closely cost, schedule, and uncertainty are tied to design choices. Even small changes in sequencing, construction

methods, or coordination requirements had direct implications for productivity, cost, and risk exposure.

This evolving understanding led to a more holistic design approach, one in which structural systems, construction sequencing, logistics, cost, and risk were evaluated together rather than in isolation. Design decisions were assessed not only for technical adequacy but also for their effects on site operations, safety, and schedule. The resulting final design reflects a coordinated strategy that prioritizes constructability within a highly constrained campus environment. Ultimately, the project demonstrates that successful delivery depends on aligning design with real-world conditions, ensuring that underpinning, tunnel construction, logistics, and trade coordination function as interconnected parts of a cohesive system.

### **Design Constraints**

The McIntire School Expansion project was governed by a range of project-specific constraints that directly influenced design decisions, construction sequencing, and overall project execution. These constraints were not considered in isolation; instead, they were evaluated collectively to develop solutions that balanced safety, constructability, cost, and schedule.

### **Existing Structure Constraints**

The renovation of Cobb Hall introduced significant uncertainty due to limited as-built documentation and the condition of the existing structure. Existing columns were required to be removed to create larger open spaces, and the original structural system was not fully known prior to construction. This constraint directly impacted the design of underpinning and structural reinforcement systems. As a result, the team incorporated conservative design assumptions, allowed for flexibility in beam sizing, and accounted for several design options during construction. This constraint also influenced sequencing, requiring careful coordination between demolition, temporary support, and structural installation to maintain building stability.

### **Underpinning and Support-of-Excavation Constraints**

The need to excavate adjacent to an existing structure imposed strict limitations on allowable ground movement and structural settlement. The presence of ML soils and high groundwater conditions further limited the range of feasible support systems. These constraints directly influenced the selection of a secant pile wall system, which provided the stiffness and

groundwater control necessary to protect Cobb Hall. The design also required staged excavation to ensure performance criteria were met throughout construction.

### **Subsurface Utility Constraints and Tunnel Alignment**

The site contained a dense network of existing utilities, including high-voltage electrical ductbanks, chilled water lines, sanitary sewer, and domestic water systems. These utilities conflicted with the proposed tunnel alignment and required relocation or protection prior to excavation. This constraint significantly impacted both design and sequencing, as utility relocations had to be completed before tunnel construction could begin. The team addressed this by incorporating 3D coordination, utility mapping based on ASCE 38 standards, and phased relocation plans into the sequencing strategy.

### **Site Logistics and Space Constraints**

The project site was extremely constrained, with limited space for material laydown, equipment staging, and construction access. In addition, the site was located within an active campus environment with high pedestrian and vehicular traffic, including proximity to UVA Health facilities; while proximity to UVA Health Facilities was considered, it had no significant impact to the outlines scope of work. These constraints directly impacted the site logistics plan, requiring the use of just-in-time delivery, multiple controlled access points, and temporary traffic control measures such as flaggers and coned-off lanes. The limited space also influenced crane placement, routing of construction vehicles, and overall phasing of work.

### **Schedule Constraint**

The requirement to complete Cobb and Shumway Halls in time for the Fall 2025 semester imposed a fixed and non-negotiable deadline. This constraint affected all aspects of design and planning, requiring the development of a detailed construction sequence, identification of critical path activities, and inclusion of schedule contingencies. Decisions related to construction methods, such as the selection of underpinning systems and sequencing of utility work, were evaluated based on their ability to support timely project delivery.

### **Material Availability and Market Constraints**

The project was developed during a period of high market demand, resulting in limited subcontractor availability and potential delays in procuring long-lead materials such as mechanical equipment, specialty finishes, and lighting systems. These constraints influenced procurement strategies, requiring early buyout of key trades, identification of alternate suppliers, and incorporation of flexibility into the project schedule to account for potential delays.

### **Constructability and Trade Coordination Constraints**

The integration of new construction with an existing building, combined with complex architectural features and dense MEP systems, created significant constructability challenges. Above-ceiling spaces were particularly constrained, requiring careful coordination of mechanical, electrical, plumbing, and fire protection systems. This constraint directly influenced the use of BIM and clash detection to resolve conflicts prior to construction, as well as the sequencing of trades to minimize congestion and rework.

### **Safety and Code Compliance Constraints**

Maintaining a safe environment for construction workers, students, faculty, and the surrounding community was a primary constraint throughout the project. This included compliance with VUSBC requirements, as well as the implementation of site-specific safety measures such as pedestrian detours, exclusion zones, and controlled access points. Safety considerations also influenced crane operations, excavation support design, and overall site logistics to minimize risk in a highly active environment.

### **Client and Stakeholder Constraints**

The University of Virginia required the project to maintain campus operations, minimize disruption, and achieve a high level of architectural quality consistent with the surrounding campus. This included detailed finish requirements, integration with existing buildings, and preservation of campus aesthetics. These constraints influenced material selection, construction sequencing, and quality control measures, requiring mockups, detailed coordination, and careful execution of architectural elements.

### **Summary of Impact on Design**



These constraints collectively shaped the project's design and execution strategy. Decisions related to underpinning, tunnel alignment, site logistics, and construction sequencing were driven not only by technical feasibility but also by real-world limitations such as space, safety, schedule, and coordination. By addressing these constraints early and incorporating them into the design process, the team developed a solution that is both technically sound and constructible within the unique conditions of the McIntire School Expansion project.

## DESIGN

### Support of Excavation Analysis

Our team evaluated several underpinning and support-of-excavation systems for the existing foundations at Cobb Hall and developed a preliminary structural design for the option that best fits the site. The work is guided by a mix of geotechnical constraints, building loads from Cobb and Shumway, groundwater conditions, and the construction limits of the dense UVA campus environment. To stay aligned with industry practice, we are using ASCE 7-22 to establish vertical and lateral load demands, the 2021 IBC as adopted by the Virginia USBC for structural design requirements, and FHWA/ASCE guidance documents for drilled shafts and temporary support systems. All subsurface parameters come from the geotechnical borings closest to the planned excavation, which identify the soils as ML sandy silts with high fines and a relatively shallow groundwater table.

The first step in our process is establishing how much load the existing building foundations will transfer into the temporary support system. We identified the controlling live and dead loads at the foundation elevation and applied the LRFD load combinations from ASCE 7 to determine the factored vertical demand. This gives us the required bearing capacity that the underpinning elements must provide throughout excavation. At the same time, we determined the lateral pressures that will act on the support-of-excavation system. These pressures define the bending, shear, and deformation limits that the wall must be designed to resist.

Using these load and pressure calculations, we then evaluated several underpinning and support-of-excavation options, as shown in Figures 1-3. Performing a feasibility and constructability overview for the three options, Figure 4 summarizes the cost and schedule analysis for each option, tabulating each type of support to highlight the analysis process



**Figure 1.** *Secant Pile Wall*



**Figure 2.** *Soil Nailing*



**Figure 3.** *Slurry Wall*

| Method                                             | Secant Pile Wall                                                                                                                                                                                           | Soil Nailing                                                                                                                                                                     | Slurry Wall                                                                                                                                                                            |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                                 | Drill overlapping primary and secondary piles so the reinforced secondary piles cut into the primaries, creating a continuous retaining wall.                                                              | Drill into the soil, insert reinforcing steel and grout, allow to cure, cover with shotcrete, and cap the ends of the reinforcing steel.                                         | Excavate deep trenches in stages, fill them with slurry to support the walls, insert the casing, and displace the slurry with concrete.                                                |
| <b>Cost per SF</b>                                 | \$100-125                                                                                                                                                                                                  | \$37-56                                                                                                                                                                          | \$150-200                                                                                                                                                                              |
| <b>Schedule</b>                                    | 150-300 SF/day                                                                                                                                                                                             | 300-600 SF/day                                                                                                                                                                   | 100-250 SF/day                                                                                                                                                                         |
| <b>Performance in tight utility corridors</b>      | Excellent — small equipment, drilled installation, minimal vibration                                                                                                                                       | Poor — requires open excavation and soil continuity                                                                                                                              | Poor — requires large trenching equipment and slurry handling                                                                                                                          |
| <b>Settlement control near existing structures</b> | Excellent — stiff, continuous wall                                                                                                                                                                         | Weak — relies on soil arching; not suitable for sensitive structures                                                                                                             | Good — but requires large laydown and specialized equipment                                                                                                                            |
| <b>Groundwater / watertightness</b>                | Good — interlocking piles reduce seepage                                                                                                                                                                   | Poor — shotcrete facing is not watertight                                                                                                                                        | Good — but joints can leak; slurry management required                                                                                                                                 |
| <b>Site access and equipment footprint</b>         | Small footprint; feasible in constrained sites                                                                                                                                                             | Small footprint                                                                                                                                                                  | Very large footprint; not feasible in tight areas                                                                                                                                      |
| <b>Integration with structural system</b>          | Strong — can serve as a permanent structure                                                                                                                                                                | Weak — temporary system only                                                                                                                                                     | Strong — but only economical when part of a foundation                                                                                                                                 |
| <b>Additional Considerations</b>                   | <ul style="list-style-type: none"> <li>No need for underpinning adjacent structures outside of Shumway</li> <li>Must remain completely vertical</li> <li>Must finish the wall before excavation</li> </ul> | <ul style="list-style-type: none"> <li>Not good for fine-grained soil</li> <li>Excavation and support can happen simultaneously</li> <li>Relies on careful sequencing</li> </ul> | <ul style="list-style-type: none"> <li>Flexible dimensions</li> <li>High slurry costs (manufacturing, containment, and disposal)</li> <li>Limited overlap with construction</li> </ul> |
| <b>% Effectiveness<sup>1</sup></b>                 | 90.57%                                                                                                                                                                                                     | 61.71%                                                                                                                                                                           | 47.57%                                                                                                                                                                                 |

**Figure 4. Cost and Schedule Analysis for SOE and Underpinning Options**

<sup>1</sup> Constraints labeled green is 100% effectiveness, yellow is 67%, and red is 33%; Only the average of these % effectiveness values was listed

Effectiveness scores were assigned by evaluating how well each support-of-excavation option satisfies the project's key constraints. Criteria that are fully or best met (in regards to cost and schedule) receive 100%, partially met criteria receive 67%, and criteria that are poorly met or incompatible with site conditions receive 33%. This provides a consistent, quantitative basis for comparing alternatives. The selection of a support-of-excavation system for Cobb Hall was guided by several key criteria: the ability to maintain stability in weak, cohesionless soils; control of groundwater inflow; minimization of ground deformation near sensitive existing foundations; compatibility with dense utility corridors; and the need to work within a highly constrained site footprint. Each system was evaluated against these constraints using percentage-based effectiveness scores that reflect how fully the method satisfies the project requirements.

Through this evaluation, the secant pile wall emerged as the most appropriate system for the site. The subsurface profile indicates that the native soils are unlikely to remain stable during open excavation, and the high groundwater table significantly limits the feasibility of more conventional systems such as soldier piles with lagging or soil nailing. A secant pile wall provides continuous, rigid support that maintains excavation stability, offers reliable groundwater control through interlocking piles, and minimizes deformation adjacent to existing foundations. It also accommodates the tight site geometry and dense utility network while requiring relatively small equipment and minimal laydown space- an important advantage in the constrained campus environment. Taken together, these factors make the secant pile wall the only option that fully satisfies the project's technical, environmental, and logistical constraints.

## Underpinning Design Process

Following the decision for the support of the excavation method, notes were taken from the structural drawings to identify which locations and loads needed to be accounted for. Our design team reviewed the geotechnical report and further guidance from the Schnabel team. From these reports, we determined the following characteristics of the site's soil makeup: soil classification, lateral pressure coefficient, equivalent fluid unit weight, base friction coefficient, recommendations of height and moist soil unit weight, and relevant live/dead loads on Cobb Hall. Using these characteristics documented in Appendix F, the active and passive lateral loads were calculated with the following formula:

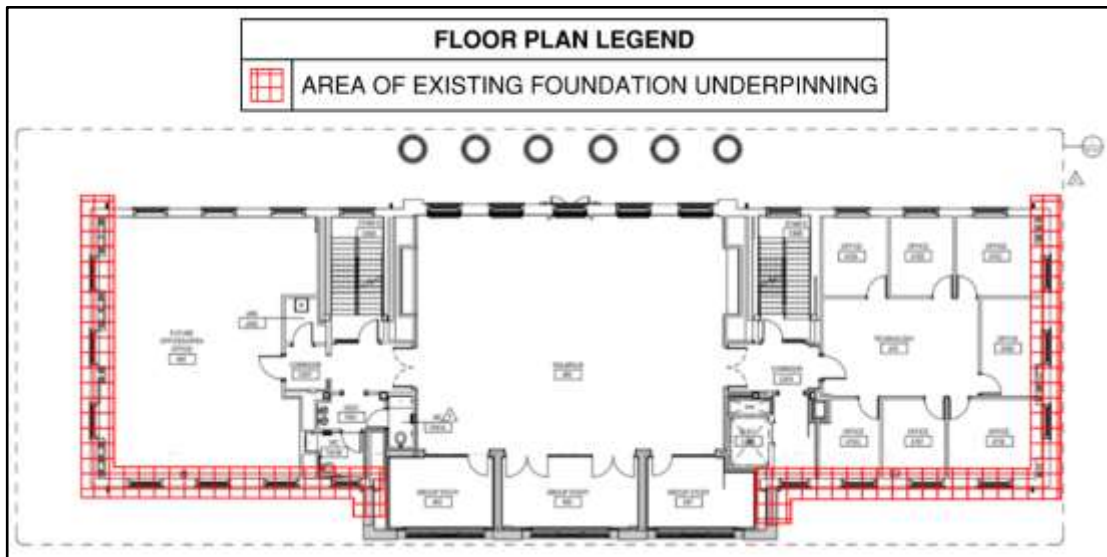
**Equation 1: Active/Passive Lateral Loads Due to Soil**

$$\text{Lateral Pressure} = K \times \gamma \times z$$

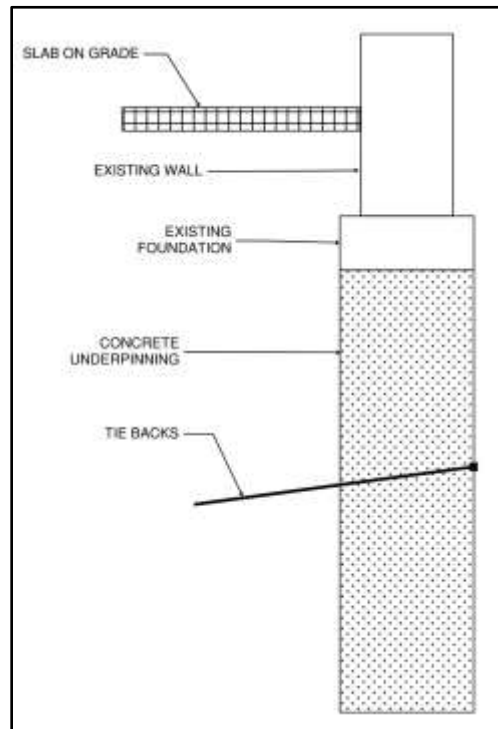
$$\text{Lateral Pressure (Active)} = 0.36 \times 120 \times 30 = 1,296 \text{ psf}$$

$$\text{Lateral Pressure (Passive)} = 0.53 \times 120 \times 30 = 1,908 \text{ psf}$$

From the structural drawings and reports, the live and dead vertical loads were determined to be 285 psf and 65 psf, respectively. Using the calculated lateral pressures and vertical loading conditions, the secant pile wall and underpinning system were further developed and detailed. The design is illustrated in Figures 5 and 6, which depict the pile layout, spacing, embedment depth, and structural configuration required to support the existing foundations. These drawings reflect the integration of geotechnical parameters, loading conditions, and constructability considerations to ensure adequate lateral support, groundwater control, and protection of the historic structure during excavation.



**Figure 5.** Cobb Hall Underpinning Drawing: Plan View



**Figure 6.** *Cobb Hall Underpinning Drawing: Typical Section*

## Tunnel Design Considerations and Codes

Construction sequencing for the tunnel must account for the dense network of existing utilities, limited site access, and the structural and safety requirements established by governing codes. The sequence must maintain continuous operation of critical utilities, provide safe excavation conditions, and allow for controlled installation of precast tunnel sections.

### Utility Constraints and Protection Requirements

Existing underground utilities impose the primary constraints on excavation and installation activities. ASCE 38-22 establishes procedures for locating, verifying, and classifying utilities, which inform allowable clearances and protection requirements established by the utility owners. Electrical and mechanical utilities must meet the separation and safety requirements of NFPA 70 and NFPA 90A, which guide temporary support, isolation, and spacing during construction. Sequencing must ensure that no utility is taken offline without an approved temporary arrangement and that all required horizontal and vertical separations are maintained throughout excavation and placement.

## **Excavation and Support-of-Excavation Requirements**

Excavation activities must comply with the structural and safety provisions of the Virginia Uniform Statewide Building Code (USBC), which incorporates the International Building Code (IBC) for underground construction. Soil pressures, surcharge loads, and seismic forces are determined using ASCE 7-22 and inform the design and installation order of support-of-excavation (SOE) systems. The sequencing must ensure that excavation proceeds in controlled stages, with temporary bracing or shoring installed before advancing to deeper cuts. SOE elements must remain in place until the precast tunnel sections and surrounding backfill provide adequate structural support.

## **Precast Tunnel Placement**

As specified by the design team, the tunnel will be installed using precast sections that require coordinated sequencing to manage limited laydown space and restricted equipment access. Each section must be placed in a controlled order that maintains excavation stability and avoids extensive interference with active utilities. The sequence must also account for required bedding, alignment tolerances, and joint sealing procedures. Because precast components reduce the duration of open excavation, their placement order directly affects utility protection and SOE performance.

## **Temporary Relocations and Site Access**

Where utilities conflict with the tunnel alignment, temporary shutdowns or bypass arrangements must occur before installations happen in those areas. These shutdowns must follow the clearance and safety requirements of ASCE 38-22 and NFPA standards. Site access constraints require sequencing that minimizes equipment congestion and maintains safe pedestrian and vehicular circulation around the work zone. Deliveries of precast components must be timed to avoid extended on-site storage and to maintain unobstructed access for excavation and SOE operations.

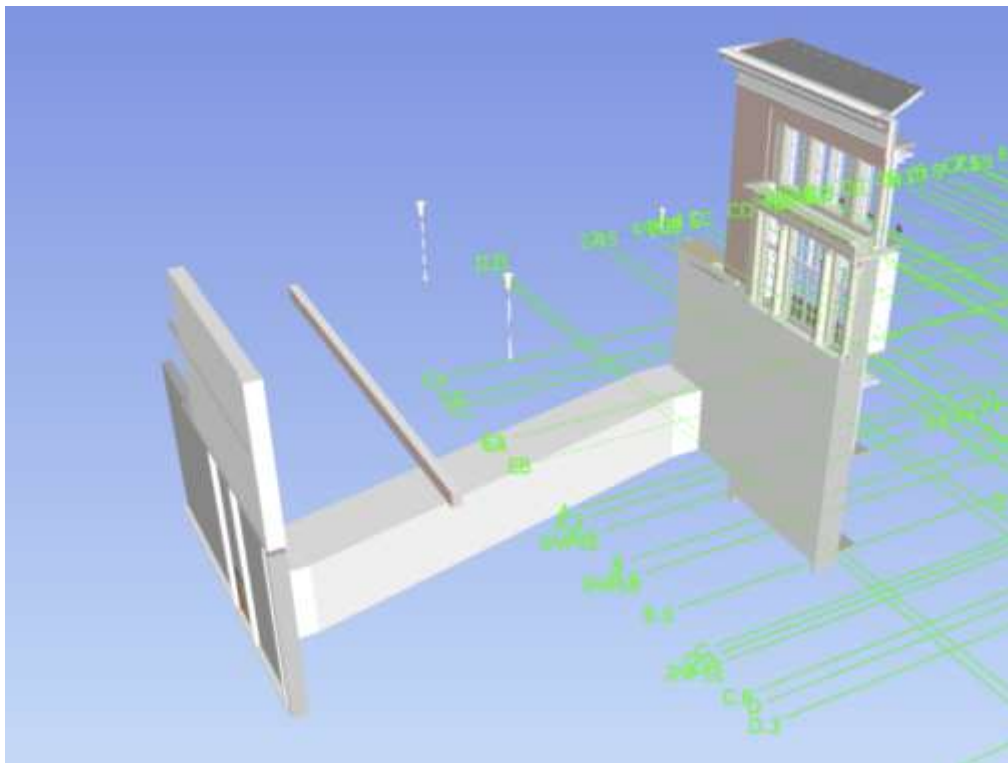
## **Environmental and Operational Continuity**

Because the work occurs adjacent to active academic and mechanical facilities, sequencing must limit vibration, noise, and service interruptions. Critical systems - such as

chilled water, electrical feeders, and communication lines - must remain operational throughout construction, or communicated to serviced facilities elsewhere. Sequencing must therefore prioritize protection and staged utility shutdowns before excavation and installment approaches their corridors.

## Tunnel Sequencing Planning

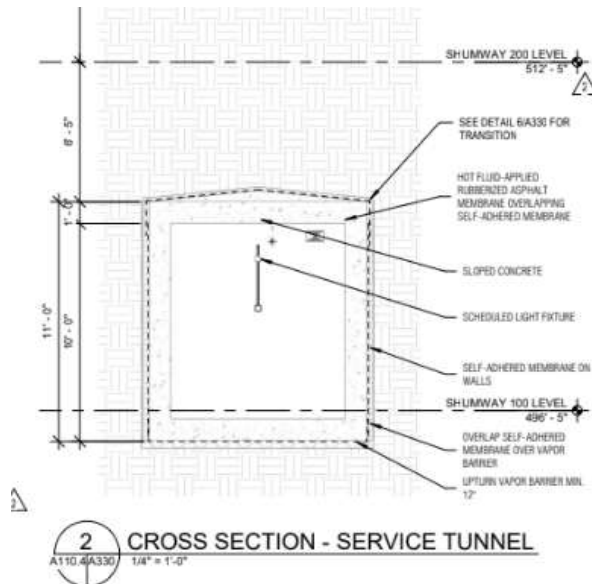
Developing an appropriate construction sequence requires a workflow that accounts for existing and proposed utilities, structural connection points between facilities, and the movement of materials and equipment within a constrained site. Establishing the tunnel's dimensional requirements is the first governing constraint, as projected by Figure 7.



**Figure 7.** *3D Navisworks Model of Tunnel Structure*

Based on the design drawings, two top-of-tunnel elevation conditions must be satisfied. Using the controlling cases from each condition (top of tunnel being either 506' or 505' - 2"), the tunnel must be installed between elevations 494'-2" and 506'-0" to maintain the required 11-ft internal design height while accommodating both top-of-tunnel elevations shown in Figure 8. This elevation range defines the allowable vertical envelope for excavation, support - of - excavation systems, and utility clearances during construction.





**Figure 8.** *Tunnel Dimensions & Elevations (Top: Sheet A330; Bottom: Sheet C101)*

Underground utility clash detection confirmed that all new utilities are fully coordinated across the civil, structural, and architectural drawings, as documented in Appendix E. For existing utilities beneath the plaza, verification procedures described in Appendix F established that each utility was accurately located, evaluated against the tunnel alignment, and found to be compatible with the proposed design. As a result, no permanent relocations are required, and all existing utilities can remain in service with standard protection measures during construction.

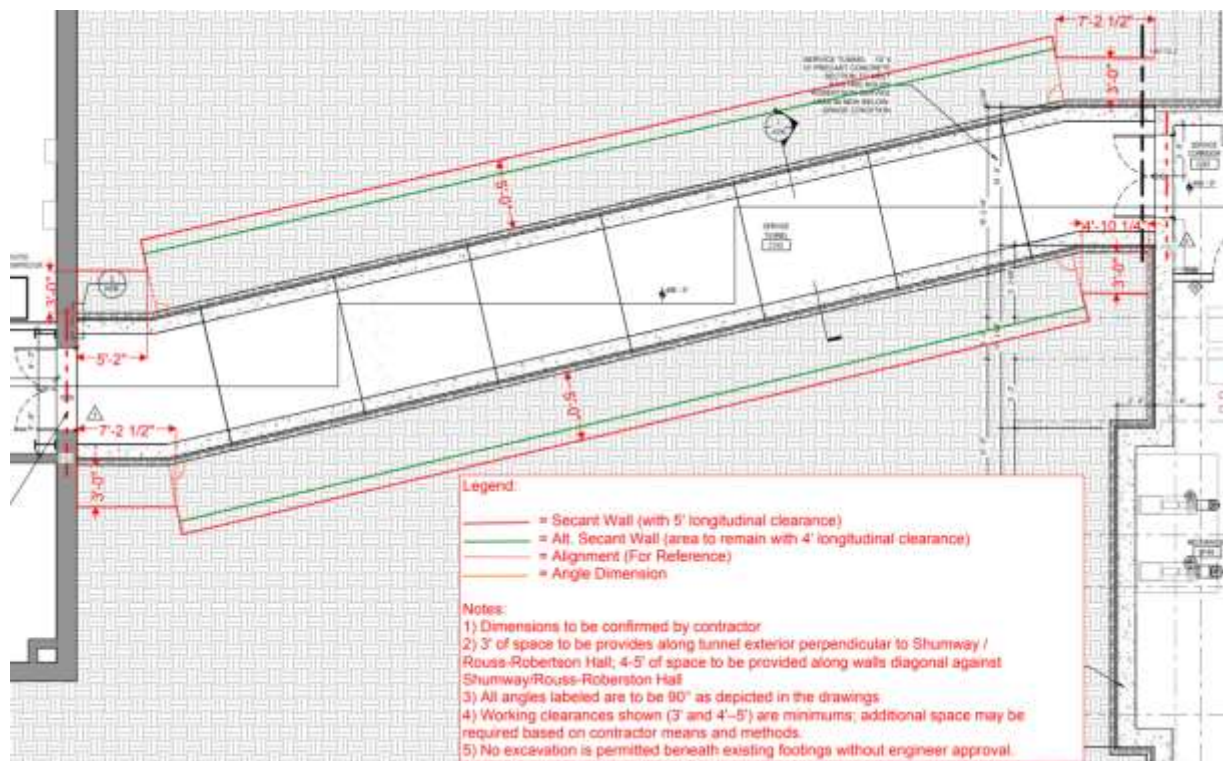
### **Phase 1: Installation of Support of Excavation**

Secant pile walls are installed along the longitudinal edges of the tunnel alignment after underpinning is completed and the backfill is brought to grade. Before drilling begins, all existing utilities within the SOE influence zone are exposed, verified, and protected in accordance with the utility investigation procedures described in ASCE 38-22. Where utilities fall within the anticipated secant pile footprint or drilling tolerance envelope, the contractor must excavate around them, install temporary supports as needed, and encase each line in concrete or

controlled-density fill to prevent damage from drilling vibrations, ground loss, or casing oscillation.

The secant wall alignment is positioned approximately 4–5 feet outside the exterior face of the tunnel to maintain working clearance for excavation, waterproofing, and placement of precast sections. Drilling proceeds in alternating primary and secondary piles to achieve interlock, followed by reinforcement and concrete placement. Protected utilities remain in service throughout this process, with monitoring performed until SOE installation is complete.

At each building connection, the standard SOE offset is locally modified to allow direct access to the building foundation wall. Short transverse secant wall segments are installed to form a small excavation “pocket” that reaches the building face, providing a stable, fully supported zone for exterior waterproofing and joint detailing at the tunnel–building interface. Once all piles are installed, the tops are trimmed to design elevation and any required capping beams or alignment guides are placed. See Figure 9 for the secant wall configuration.



**Figure 9. Secant Wall Placement Plan**

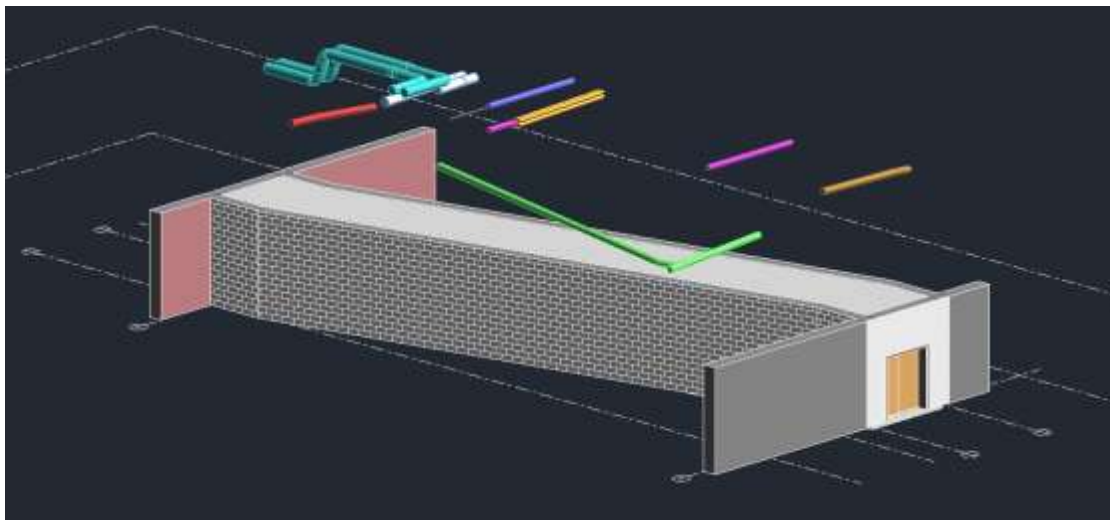
## Phase 2: Excavation

Excavation proceeds in controlled lifts to maintain stability and limit ground movement adjacent to existing foundations. Soil is removed between the longitudinal secant walls along the

tunnel alignment, with continuous monitoring for wall deflection and installation of temporary bracing or internal struts if required by the design. As excavation approaches each building, the localized SOE pockets allow the trench to extend fully to the building foundation wall. This provides unobstructed access for waterproofing continuity and joint preparation. Excavation continues to the tunnel subgrade, with careful coordination around known utilities and any required potholing or verification. The base is then fine-graded and prepared for tunnel placement, as specified by Schnabel Engineering.

### Phase 3: Tunnel Placement & Connections

The Revit-generated 3D model of the tunnel and surrounding utilities (new and existing), shown in Figure 10, serves as the primary coordination tool for placement sequencing. This model establishes the spatial relationship between the precast tunnel sections and all utility crossings, allowing the contractor to plan lifting paths, verify clearances, and identify the specific locations where temporary utility shutdowns are required.



| Utility ID | Utility           | Diameter (in) | Color  |
|------------|-------------------|---------------|--------|
| NE         | New Electrical    | 4             | Yellow |
| NCW        | New Chilled Water | 14 to 12      | Cyan   |

|     |                        |               |               |
|-----|------------------------|---------------|---------------|
| NSD | New Storm Drain        | 8             | Orange        |
| NUD | New Underdrain         | 6             | Magenta       |
| ESD | Existing Storm Drain   | 8             | Red           |
| ECW | Existing Chilled Water | 12            | White w/ Blue |
| EE  | Existing Electrical    | 2 (wire only) | Gray          |
| EW  | Existing Water         | 6             | Navy Blue     |
| ES  | Existing Sewer         | 8             | Green         |

**Figure 10.** *3D Revit Model of Tunnel with Utilities (top) and Legend (bottom)*

Once excavation reaches design grade and the SOE is fully installed, precast tunnel sections are delivered to the site and lifted into the excavation using the designated crane access corridor. Placement proceeds sequentially along the alignment, following the order established in the 3D model to avoid conflicts with active utilities and to maintain safe working distances from protected lines, which is projected into a plan view as shown in Appendix G. At each utility crossing, the model-based coordination ensures that shutdowns are limited to the minimum duration necessary and that all affected facilities are notified in advance. See Figure 11 for a specific list of tunnel sections to be installed as well as appropriate utilities to be shut down as necessary (see Appendix G for tunnel section references):

| Step | Activity                     | Utilities to be closed                | Additional Utilities at Risk/Notes                                                                                                   |
|------|------------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Install Tunnel (7 and 8)     | N/A                                   | Close proximity to ES; maintain and protect                                                                                          |
| 2    | Install tunnel (5 and 6)     | ES                                    | Close proximity to EW; maintain and protect                                                                                          |
| 3    | Install tunnel (2, 3, and 4) | ES, EE, EW, ECW 1, ECW 2 <sup>2</sup> | N/A; Greatest risk requiring careful coordination with campus activity due to several essential utilities temporarily out of service |
| 4    | Install tunnel (1)           | ESD                                   | N/A; coordinate as a weather related activity                                                                                        |
| 5    | Waterproofing and Inspection | N/A                                   | All; Low risk activity                                                                                                               |

**Figure 11.** *Tunnel Sequencing Plan with Utilities to be Out*

After each section is set, crews complete joint alignment, bedding adjustments, and temporary stabilization before advancing to the next placement. This model-driven approach

<sup>2</sup> ECW1 refers to chilled water line closest to Robertson Hall and ECW 2 refers to CW line closest to Cobb/Shumway Hall

minimizes open-cut exposure, maintains continuity of critical campus utilities, and ensures that tunnel placement remains fully coordinated with the protection measures established during SOE installation.

#### **Phase 4: New Utility Installation**

With the tunnel structure in place, new utilities are installed within the soil backfill above the tunnel roof in accordance with the coordinated utility plan. Each utility is placed at its designated elevation to maintain required cover, separation, and protection. Bedding material, as specified by the geotechnical engineer, is placed and compacted around pipes and conduits to prevent point loading, and utilities that require additional protection - such as shallow water or electrical lines - are installed with concrete encasement or protective slabs as specified. Where utilities cross perpendicular to the tunnel, they are graded to maintain clearance above the tunnel roof and avoid conflicts with the SOE. Temporary utility outages will be required for installing chilled water and electrical tie-ins to existing utility lines. All installations are inspected, pressure-tested, or continuity-tested as applicable, and verified for proper cover before final backfilling begins.

#### **Phase 5: Backfill and Grading**

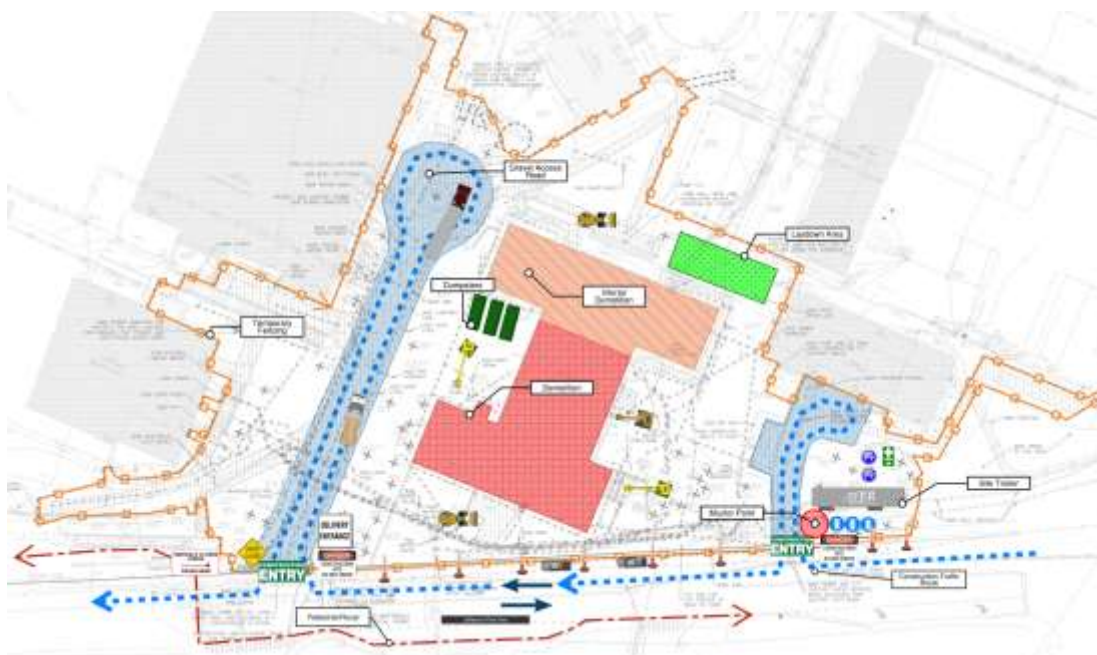
Backfilling begins once the tunnel and utilities are secured. Material is placed in controlled lifts and compacted to prevent settlement and protect the tunnel structure, as specified by the geotechnical report. Backfill around the secant wall is placed carefully to avoid overstressing the piles, and any drainage layers or waterproofing protection boards are installed as required. The final grade is restored to match site contours, and surface improvements - such as pavements, landscaping, or hardscape - are reconstructed to complete the work.

### **Site Logistics Plan**

A comprehensive site logistics plan was developed to effectively manage the constrained campus environment at UVA and ensure safe, efficient construction operations throughout the Cobb Hall expansion and renovation. Given the project's location within an active academic setting, the plan prioritizes minimizing disruption to ongoing university activities while maintaining clear and safe circulation for students, faculty, and staff.

## **Phase 1: Demolition and Excavation**

The Phase 1 demolition and excavation site logistics plan was developed to safely and efficiently manage construction operations within the constrained campus environment while maintaining clear separation between construction and pedestrian activity. The plan incorporates two designated construction access points to improve traffic flow and reduce congestion, with a defined internal haul route that supports continuous movement of equipment and dump trucks throughout the site. Unlike earlier iterations (see Appendix B), where construction access roads were too narrow to safely accommodate large equipment and tractor-trailers, this revised plan provides adequately sized routes to ensure safe maneuverability and operational efficiency. The existing roadway within the site footprint is planned for removal to accommodate demolition and excavation activities and will be replaced with a temporary gravel access road to maintain site circulation and provide a stable surface for heavy construction traffic. Demolition activities are clearly separated between full structural demolition and interior selective demolition zones, with adjacent laydown areas and dumpsters positioned to support efficient debris handling and removal. Temporary fencing secures the perimeter, while designated pedestrian detours and signage maintain safe campus circulation. Additional features such as equipment exclusion zones, a centralized muster point, and strategically located laydown areas further enhance safety and coordination. Overall, the plan reflected in Figure 12 is a refined approach that improves constructability, site access, and safety compared to previous iterations while minimizing disruption to the surrounding university environment.



**Figure 12.** *Demolition and Excavation Site Logistics Plan*

## Phase 2: Foundation and Structural Construction

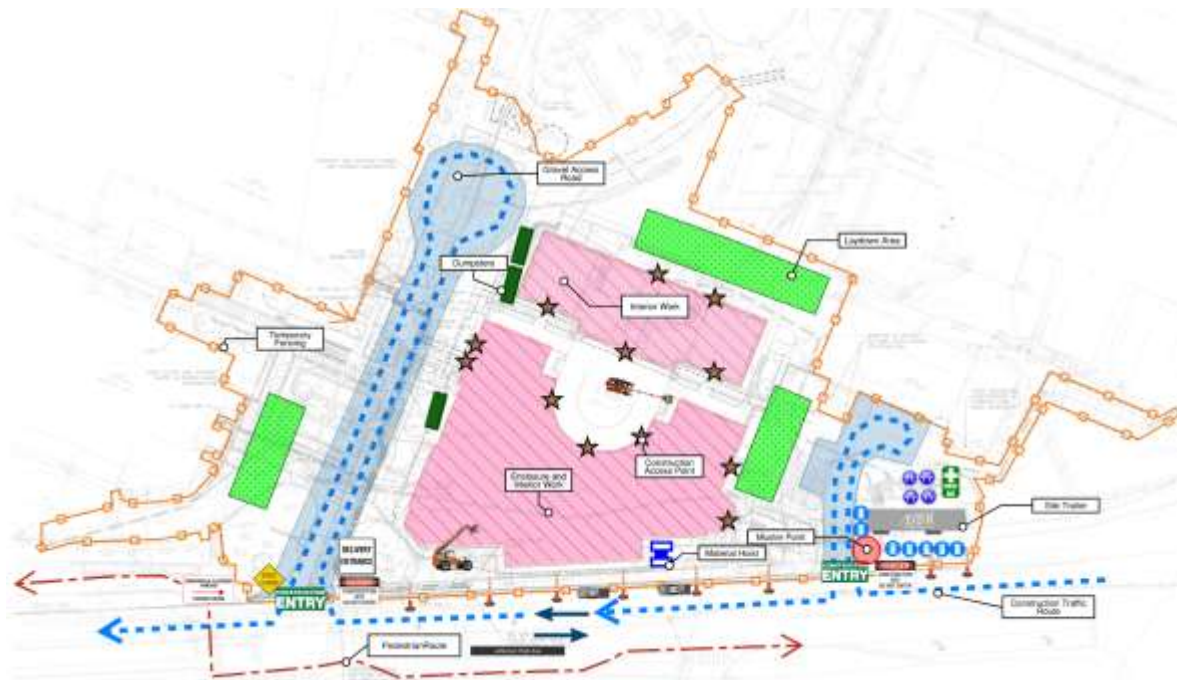
The Phase 2 site logistics plan reflects the transition to foundation and structural construction, with a focus on safe coordination of vertical building operations within a highly constrained campus environment. A centrally located crane is introduced to support the erection of the new Shumway Hall addition, with its swing radius of 164 feet carefully mapped to optimize coverage of the building footprint and adjacent laydown areas. The crane is erected at the beginning of Phase 2 to support structural operations and is planned to remain active through the completion of the primary framing scope. Due to the crane's reach extending over portions of the site circulation network, the primary gravel access road beneath the swing radius is designated as limited access and is subject to controlled entry. All crane swing movements extending beyond the immediate work zone are coordinated in advance with safety planners, stakeholders, and site personnel to ensure controlled operations and minimize risk. During active overhead lifting operations, traffic control personnel will temporarily halt vehicle movement along this route to ensure safety, and major lifts will be scheduled during off-peak hours to minimize disruption to surrounding campus and hospital circulation. As construction progresses, an expanded laydown area is established on the north end of Cobb Hall to support increased material demands associated with structural framing. Given the limited available space, the plan







continues to rely on just-in-time delivery strategies to minimize on-site storage and prevent overcrowding. The temporary gravel access road remains in use to support construction traffic, with defined delivery routes to maintain circulation. Pedestrian pathways are preserved and clearly separated from construction operations through fencing, signage, and rerouting to ensure safety within the active campus environment. Additional safety measures, including a centralized muster point and controlled site entry points, support safe operations during this highly congested phase. The plan in Figure 14 demonstrates a coordinated approach to managing overlapping trades while maintaining safety, efficiency, and minimal disruption to surrounding campus operations.

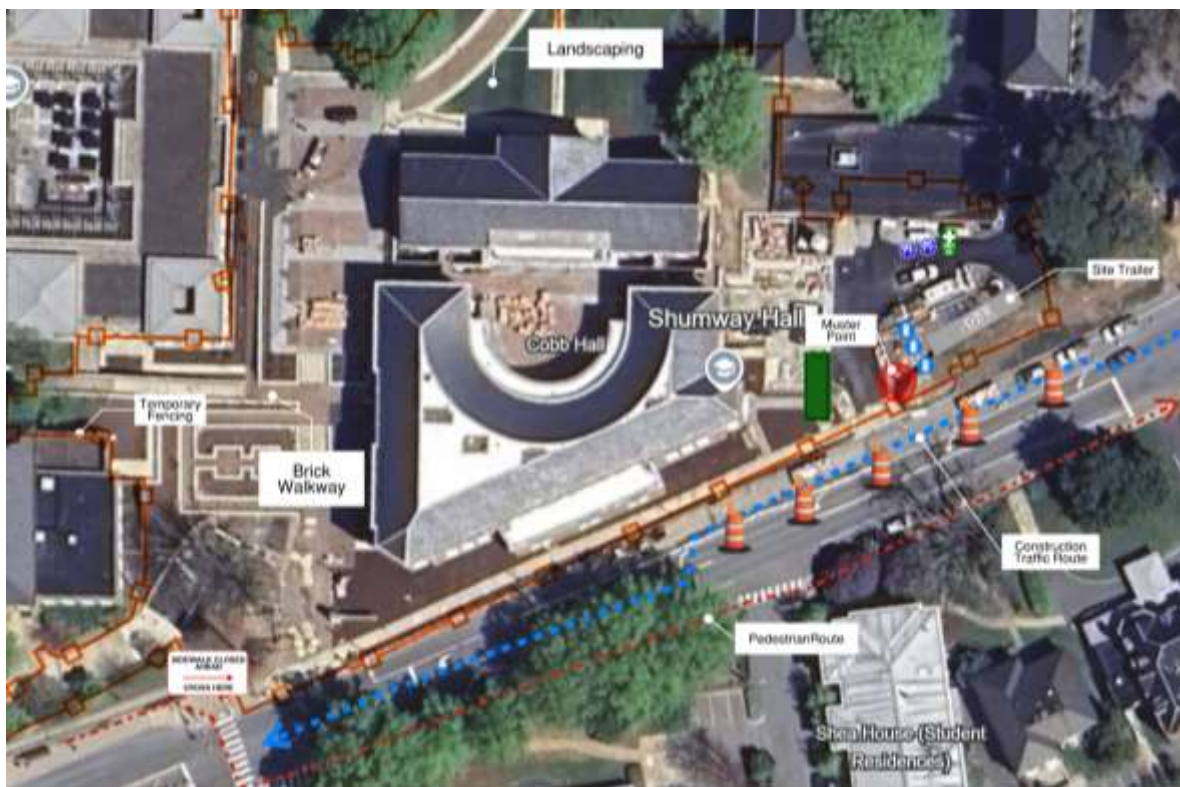


**Figure 14.** *Enclosure and Interior Phase Site Logistics Plan*

#### **Phase 4: Closeout and Site Restoration**

The Phase 4 site logistics plan represents project closeout and final site restoration, in which construction activities are significantly reduced but not yet fully complete. Temporary construction elements, including the material hoist and most laydown areas, are removed to clear the site, while remaining fencing and controlled access points are maintained to support ongoing punch list work and final inspections. Installation of permanent site features, including brick walkways, outdoor stair connections, and landscaped lawn areas, is completed during this phase,

replacing the temporary gravel access routes used in earlier stages. Access to the site remains restricted to authorized personnel while final work is completed. Vehicular activity is limited to necessary service and delivery operations, with construction vehicles accessing the site through a temporarily coned-off lane along Jefferson Park Avenue to allow for brief, controlled entry and exit. Primary traffic flow is maintained along perimeter roads, while pedestrian routes remain clearly defined and separated from these limited construction activities. Finally, this phase, as reflected in Figure 15, transitioned from active construction to a functional campus space, while maintaining controlled site conditions to ensure safety and quality through project completion.



**Figure 15.** *Closeout and Site Restoration Site Logistics Plan*

## Cast-in-Place Concrete Self-Perform

A detailed estimate was prepared for self-performing the cast-in-place concrete scope, incorporating a full quantity survey with material pricing, labor, and equipment rentals. Previous estimates were limited to just material pricing. This estimate accounts for concrete, formwork, reinforcement, placement, finishing, and curing. The final step in this estimate is analyzing labor productivity and potential constraints so the team can ultimately develop a realistic projection for

time, enabling accurate budgeting and resource allocation. This approach ensures that the concrete work can be executed efficiently while maintaining the quality standards required for both Cobb and Shumway Halls. This quantity survey involved a thorough examination of the provided construction documents in order to properly interpret notes and evaluate the scope. All estimates were prepared using Bluebeam Revu using various measurement tools. Screenshots from this exercise are found in Appendix D.

Seen in Table 1 is the quantity survey of the column footings for Cobb and Shumway halls. Notable values include the final concrete volume of 227 CY and 8,100 ft of total rebar. Formwork was calculated based on the perimeters of each column footing.

**Table 1.** Column Footings Quantity Survey

| Column Footings | Quantity | L (ft) | W (ft) | H (ft) | Vol. (CF)    | Vol. (CY)  | SF           | # of bars | Gauge | ft of bars   | Form-work    |
|-----------------|----------|--------|--------|--------|--------------|------------|--------------|-----------|-------|--------------|--------------|
| CF4             | 2        | 4      | 4      | 1.00   | 32           | 1.2        | 32           | 8         | 5     | 64           | 32           |
| CF6             | 4        | 6      | 6      | 1.00   | 144          | 5.3        | 144          | 10        | 6     | 240          | 96           |
| CF7             | 5        | 7      | 7      | 1.16   | 284          | 10.5       | 245          | 12        | 6     | 420          | 162.4        |
| CF8             | 9        | 8      | 8      | 1.33   | 766          | 28.4       | 576          | 14        | 6     | 1008         | 383.04       |
| CF9             | 1        | 9      | 9      | 1.50   | 122          | 4.5        | 81           | 12        | 7     | 108          | 54           |
| CF10            | 7        | 10     | 10     | 1.67   | 1169         | 43.3       | 700          | 16        | 7     | 1120         | 467.6        |
| CF10.5          | 1        | 10.5   | 10.5   | 2.08   | 229          | 8.5        | 110          | 20        | 7     | 210          | 87.36        |
| CF11            | 3        | 11     | 11     | 1.83   | 664          | 24.6       | 363          | 18        | 7     | 594          | 241.56       |
| CF12T           | 2        | 12     | 12     | 2.00   | 576          | 21.3       | 288          | 44        | 7     | 1056         | 192          |
| CF13            | 1        | 13     | 13     | 2.00   | 338          | 12.5       | 169          | 24        | 7     | 312          | 104          |
| CF14            | 1        | 14     | 14     | 2.16   | 423          | 15.7       | 196          | 28        | 7     | 392          | 120.96       |
| CF14T           | 2        | 14     | 14     | 2.16   | 847          | 31.4       | 392          | 56        | 7     | 1568         | 241.92       |
| CF14x12         | 0        | 14     | 12     | 2.00   | 0            | 0.0        | 0            | 29        | 7     | 0            | 0            |
| CF18x12         | 0        | 18     | 12     | 2.33   | 0            | 0.0        | 0            | 36        | 7     | 0            | 0            |
| CF22x12         | 1        | 22     | 12     | 2.00   | 528          | 19.6       | 264          | 64        | 7     | 1008         | 136          |
| <b>TOTAL</b>    |          |        |        |        | <b>6,122</b> | <b>227</b> | <b>3,560</b> |           |       | <b>8,100</b> | <b>2,319</b> |

Following the survey of column footings, we performed a quantity survey of the wall footings for Cobb and Shumway halls. WF12.5, WF13.5, and WF14 include multiple rows to account for varying types of continuous and transverse rebar. Notable values include the final concrete volume of 418 CY, as well as 26,998 ft of total rebar. Formwork was calculated based on the total perimeter of each wall footing. This can be seen in Table 2 below.

**Table 2.** Wall Footings Quantity Survey

| Wall Footings | L (ft) | W (ft) | H (ft) | CF            | CY         | SF           | Cont bars | Gauge | ft of bars    | Trans bars | Gauge | Spacing | ft of bars    | Form-work    |
|---------------|--------|--------|--------|---------------|------------|--------------|-----------|-------|---------------|------------|-------|---------|---------------|--------------|
| WF2           | 157.5  | 2      | 1.00   | 315           | 12         | 315          | 3         | 4     | 473           | 1          | 4     | 4.00    | 79            | 315          |
| WF2A          | 34.5   | 2      | 1.00   | 69            | 3          | 69           | 6         | 4     | 207           | 2          | 4     | 4.00    | 35            | 69           |
| WF3           | 0      | 3      | 1.00   | 0             | 0          | 0            | 3         | 5     | 0             | 1          | 4     | 4.00    | 0             | 0            |
| WF3.5         | 148    | 3.5    | 1.00   | 518           | 19         | 518          | 6         | 4     | 888           | 2          | 5     | 1.00    | 1036          | 296          |
| WF3A          | 0      | 3      | 1.00   | 0             | 0          | 0            | 6         | 4     | 0             | 2          | 5     | 1.00    | 0             | 0            |
| WF5           | 0      | 5      | 1.50   | 0             | 0          | 0            | 4         | 6     | 0             | 2          | 6     | 1.00    | 0             | 0            |
| WF5.5         | 50.5   | 5.5    | 1.67   | 464           | 17         | 278          | 10        | 6     | 505           | 2          | 6     | 0.67    | 829           | 169          |
| WF8.5         | 162    | 8.5    | 1.67   | 2300          | 85         | 1377         | 16        | 6     | 2592          | 2          | 6     | 0.67    | 4110          | 541          |
| WF10          | 0      | 10     | 2.00   | 0             | 0          | 0            | 20        | 6     | 0             | 2          | 6     | 1.00    | 0             | 0            |
| WF11.5        | 47     | 11.5   | 1.67   | 903           | 33         | 541          | 20        | 6     | 940           | 2          | 7     | 0.67    | 1613          | 157          |
| WF12.5        | 122    | 12.5   | 2.00   | 3050          | 113        | 1525         | 24        | 7     | 2928          | 1          | 7     | 1.00    | 1525          | 488          |
|               | 122    | 12.5   | 2.00   |               |            |              |           |       |               | 1          | 8     | 0.67    | 2276          | 0            |
| WF13.5        | 55     | 13.5   | 2.00   | 1485          | 55         | 743          | 26        | 7     | 1430          | 1          | 7     | 1.00    | 743           | 220          |
|               | 55     | 13.5   | 2.00   |               |            |              |           |       |               | 1          | 8     | 0.67    | 1108          | 0            |
| WF14          | 62.5   | 14     | 2.50   | 2188          | 81         | 875          | 24        | 6     | 1500          | 1          | 7     | 1.00    | 875           | 313          |
|               | 62.5   | 14     | 2.50   |               |            |              |           |       |               | 1          | 8     | 0.67    | 1306          | 0            |
| <b>TOTAL</b>  |        |        |        | <b>11,291</b> | <b>418</b> | <b>6,240</b> |           |       | <b>11,463</b> |            |       |         | <b>15,535</b> | <b>2,567</b> |

Table 3 shows the quantity survey of the slabs for Cobb and Shumway halls, split up by slab thickness and floor. The square footage value for the slab on grade was reduced to reflect the coverage provided by the column and wall footings. Notable values include the final concrete volume of 1,066 CY.

**Table 3.** Slab Quantity Survey

| Slab          | Surface Area (sf) | Depth (in) | Volume (CF) | Volume (CY) |
|---------------|-------------------|------------|-------------|-------------|
| Slab on Grade | 15925             | 5          | 6635        | 246         |
| 5 Level 2     | 5675              | 5          | 2365        | 88          |
| 3.5 Level 2   | 10843             | 3.5        | 3163        | 117         |
| 4.5 Level 2   | 2115              | 4.5        | 793         | 29          |
| 6.5 Level 2   | 1000              | 6.5        | 542         | 20          |
| 3.5 Level 3   | 19715             | 3.5        | 5750        | 213         |
| 4.5 Level 3   | 1695              | 4.5        | 636         | 24          |
| 6.5 Level 3   | 0                 | 6.5        | 0           | 0           |
| 3.5 Level 4   | 12590             | 3.5        | 3672        | 136         |
| 4.5 Level 4   | 1695              | 4.5        | 636         | 24          |
| 6.5 Level 4   | 0                 | 6.5        | 0           | 0           |
| 3.5 Level 5   | 12900             | 3.5        | 3763        | 139         |

|              |               |     |               |              |
|--------------|---------------|-----|---------------|--------------|
| 4.5 Level 5  | 0             | 4.5 | 0             | 0            |
| 6.5 Level 5  | 1550          | 6.5 | 840           | 31           |
| <b>TOTAL</b> | <b>85,703</b> |     | <b>28,793</b> | <b>1,066</b> |

The perimeter of each floor plate was also analyzed in order to determine the amount of formwork needed. Though a typical assumption might have resulted in the same perimeter value for each floor, the unique shape of the building, coupled with the atrium design, meant that each floor resulted in a unique perimeter value. Table 4 lists each of these perimeters below.

**Table 4.** Floor Plate Perimeter Survey

| <b>Floor</b> | <b>Perimeter (LF)</b> |
|--------------|-----------------------|
| 1            | 640                   |
| 2            | 1413                  |
| 3            | 725                   |
| 4            | 1044                  |
| 5            | 650                   |
| <b>TOTAL</b> | <b>4,472</b>          |

After completing the quantity survey, we performed a breakdown of each rebar gauge given the varying circumferences and unit weights. The total sum of all rebar is 53,519 lbs, or 26.8 tons. This can be seen in Table 5.

**Table 5.** Rebar Quantity Survey

| <b>Rebar gauge</b> | <b>Total ft</b> | <b>lb/ft</b> | <b>lb</b>     |
|--------------------|-----------------|--------------|---------------|
| #4                 | 1681            | 0.668        | 1123          |
| #5                 | 1100            | 1.043        | 1147          |
| #6                 | 10645           | 1.502        | 15988         |
| #7                 | 11124           | 2.044        | 22737         |
| #8                 | 4690            | 2.670        | 12523         |
| <b>TOTAL</b>       |                 |              | <b>53,519</b> |

Table 6 shows the final cost summary, based on pricing data from RSMeans. RSMeans is the industry leader in construction scoping and pricing, and was able to provide accurate values for material unit prices as well as labor and equipment costs. This value is captured in the “Ext. Total” column. Additionally, each line item is listed with the contractor’s overhead and profit to

provide a more accurate view of what this scope would ultimately cost. This is captured in the “Ext. Total O&P” column. Our resources were limited to the student version of RSMeans, which only provided accurate pricing information up to 2018. As a result, we have added a column that accounts for the escalation in construction costs over the past 7 years. This escalation factor was found using a construction cost index (CCI) provided by Mortenson Construction.

$$\text{Escalation Factor} = \frac{2025 \text{ Cost Index}}{2018 \text{ Cost Index}} = \frac{199.8}{130.8} = 1.527$$

Data for Charlottesville was limited, so the 2025 cost index and the 2018 cost index were both national metrics. The original pricing derived from RSMeans, however, is based on Charlottesville.

**Table 6.** Cost Estimates by Division

| Qty                                                  | Description                                                      | Unit | Mats        | Labor       | Equip | Ext. Total      | Ext. Total O&P (2018) | Ext. Total O&P (2025) |
|------------------------------------------------------|------------------------------------------------------------------|------|-------------|-------------|-------|-----------------|-----------------------|-----------------------|
| <b>FORMWORK — Division 03 11</b>                     |                                                                  |      |             |             |       |                 |                       |                       |
| 2,567                                                | Concrete forms, footing, continuous wall, plywood, 4 use         | SFCA | \$4,671.94  | \$3,953.18  |       | \$8,599.45      | \$11,115.11           | \$16,972.77           |
| 2,319                                                | Concrete forms, footing, spread, plywood, 4 use                  | SFCA | \$1,321.83  | \$4,174.20  |       | \$5,496.03      | \$7,815.03            | \$11,933.55           |
| 640                                                  | Concrete forms, slab on grade, edge, plywood, 4 use              | L.F. | \$160.00    | \$793.60    |       | \$953.60        | \$1,382.40            | \$2,110.92            |
| 3,832                                                | Concrete forms, elevated slab, edge, plywood, 4 use              | L.F. | \$4,061.92  | \$7,855.60  |       | \$11,917.52     | \$16,362.64           | \$24,985.75           |
| <b>FORMWORK SUBTOTAL</b>                             |                                                                  |      |             |             |       | <b>\$26,966</b> | <b>\$36,675</b>       | <b>\$56,003</b>       |
| <b>REINFORCING STEEL &amp; MESH — Division 03 20</b> |                                                                  |      |             |             |       |                 |                       |                       |
| 20.50                                                | Reinforcing steel, in place, footings, #4 to #7, A615, grade 60  | Ton  | \$18,794.40 | \$12,033.71 |       | \$30,828.11     | \$38,931.14           | \$59,447.85           |
| 6.30                                                 | Reinforcing steel, in place, footings, #8 to #18, A615, grade 60 | Ton  | \$5,775.84  | \$2,148.05  |       | \$7,923.89      | \$9,594.71            | \$14,651.12           |



|                             |                                                            |        |             |             |  |                 |                 |                  |
|-----------------------------|------------------------------------------------------------|--------|-------------|-------------|--|-----------------|-----------------|------------------|
| 159                         | Welded wire fabric,<br>plain, sheets, 6x6 -<br>W2.1 x W2.1 | C.S.F. | \$2,976.48  | \$3,129.12  |  | \$6,105.60      | \$8,070.84      | \$12,324.17      |
| 698                         | Welded wire fabric,<br>plain, sheets, 6x6 -<br>W2.9 x W2.9 | C.S.F. | \$16,996.30 | \$14,720.82 |  | \$31,717.12     | \$41,237.84     | \$62,970.18      |
| <b>REINFORCING SUBTOTAL</b> |                                                            |        |             |             |  | <b>\$76,574</b> | <b>\$97,834</b> | <b>\$149,393</b> |

| <b>CONCRETE — Division 03 30 / 03 31</b> |                                                                                  |      |              |             |            |                  |                  |                  |
|------------------------------------------|----------------------------------------------------------------------------------|------|--------------|-------------|------------|------------------|------------------|------------------|
| 1,711                                    | Structural concrete,<br>ready mix, 3000 psi                                      | C.Y. | \$224,637.19 |             |            | \$224,637        | \$246,914        | \$377,038.30     |
| 645                                      | Structural concrete,<br>placing, continuous<br>footing, shallow,<br>pumped       | C.Y. |              | \$9,275.10  | \$4,456.95 | \$13,732.05      | \$18,904.95      | \$28,867.86      |
| 246                                      | Structural concrete,<br>placing, slab on<br>grade, pumped, up to<br>6" thick     | C.Y. |              | \$4,071.30  | \$1,975.38 | \$6,046.68       | \$8,368.92       | \$12,779.34      |
| 821                                      | Structural concrete,<br>placing, elevated<br>slab, pumped, less<br>than 6" thick | C.Y. |              | \$12,651.61 | \$6,083.61 | \$18,735.22      | \$25,812.24      | \$39,415.29      |
| <b>CONCRETE SUBTOTAL</b>                 |                                                                                  |      |              |             |            | <b>\$263,151</b> | <b>\$300,000</b> | <b>\$458,100</b> |

|                                             |  |  |  |  |  |                  |                  |                  |
|---------------------------------------------|--|--|--|--|--|------------------|------------------|------------------|
| <b>GRAND TOTAL (Extended Total O&amp;P)</b> |  |  |  |  |  | <b>\$366,692</b> | <b>\$434,510</b> | <b>\$638,511</b> |
|---------------------------------------------|--|--|--|--|--|------------------|------------------|------------------|

The labor costs included in this estimate were developed using RSMeans data, which provides standardized productivity rates and associated labor and equipment values for each construction activity. These costs account for major operations such as formwork installation, reinforcing steel placement, and concrete placement and finishing. Self-performing this scope can provide advantages to the contractor, including improved coordination between activities, increased control over sequencing, and greater flexibility in responding to field conditions. These advantages are particularly beneficial in projects where work can be performed continuously and crews can remain fully utilized. However, self-performing also introduces potential disadvantages, as it requires maintaining a consistent workflow to ensure that labor resources are being used efficiently. If work is not continuous, crews may experience periods of reduced productivity, resulting in increased costs without corresponding progress.

Based on the conditions of the McIntire School Expansion project, maintaining this level of consistent productivity is unlikely. The project is located within a constrained campus environment with limited laydown space, restricted access points, and significant pedestrian and vehicular traffic, all of which impact crew movement and material handling. In addition, construction activities are highly dependent on sequencing, with concrete work tied to the completion of support-of-excavation systems, utility coordination, and staged excavation operations. The presence of existing utilities further introduces coordination requirements that may delay work until conditions are verified or protected. These constraints create interruptions in workflow and increase the likelihood of idle labor if the work were self-performed. As a result, subcontracting the cast-in-place concrete scope is a more effective approach, as it allows for labor to be mobilized as needed and enables the work to be completed efficiently within the project's logistical and sequencing constraints.

## Risk Register

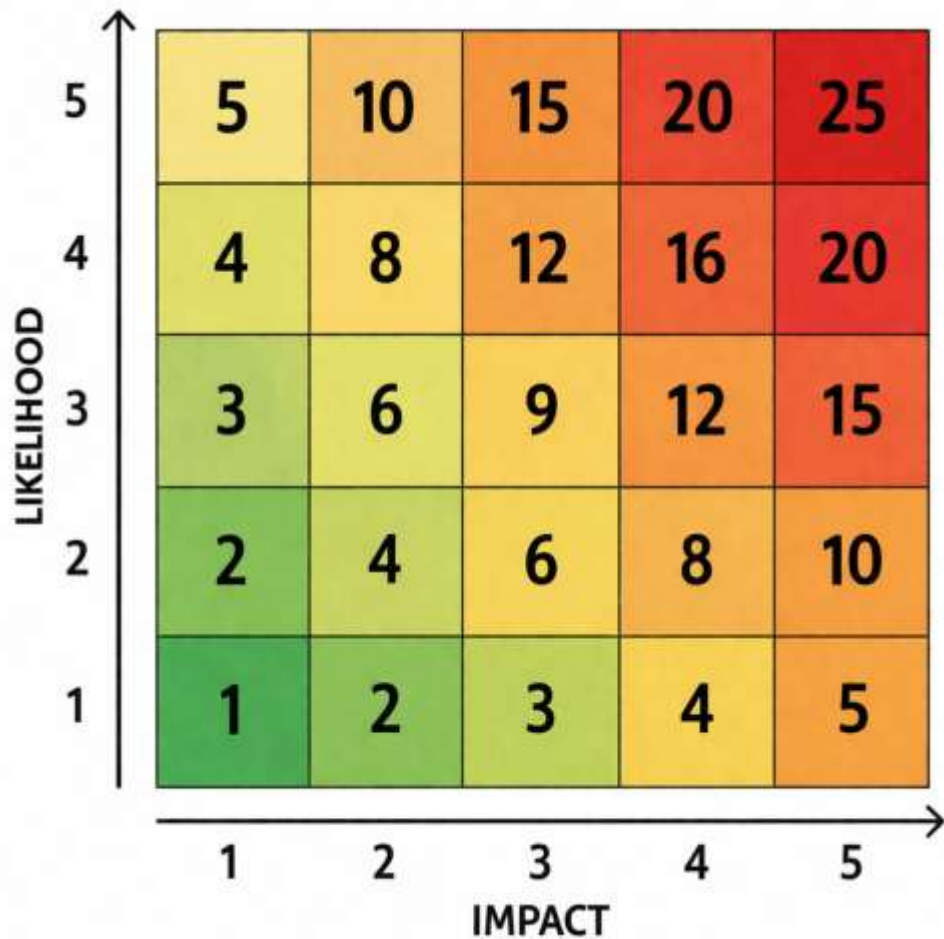
To proactively manage the challenges of the McIntire Expansion, a comprehensive risk register has been developed, as shown in Figure 14. This document identifies the major risks that could impact the project, including schedule delays, cost escalations, constructability challenges, safety concerns, and conflicts with existing utilities. Each risk is clearly defined and tied to key aspects of the project, such as underpinning of the existing structure, tunnel construction, site logistics, and coordination between multiple trades in a constrained campus environment.

To better understand and prioritize these risks, a risk matrix, as seen in Figure 14, is used to evaluate each one based on its likelihood of occurrence and potential impact. Likelihood represents how probable it is that a risk will occur, while impact reflects the severity of its effect on the project, including schedule, cost, safety, and quality. By combining these two factors, each risk is assigned a level of priority, allowing the team to focus attention on the most critical risks that could significantly affect project performance.

Based on this evaluation, mitigation strategies are developed for each identified risk to reduce or manage potential impacts. These strategies include early coordination, detailed planning, phased construction approaches, and proactive monitoring throughout the project. By identifying risks early and planning accordingly, the team is better equipped to anticipate



challenges and respond effectively, supporting the successful delivery of Cobb and Shumway Halls on schedule, within budget, and to the required quality standards.



**Figure 16.** *Risk Matrix*

| Risk ID | Risk Description                                                                                                                                   | Likelihood | Impact | Risk Score | Mitigation Strategy                                                                                                                                                                                                                  |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | Unexpected conditions in existing Cobb Hall structure, undocumented structure leading to redesign (existing columns removal, deeper beams).        | 5          | 5      | 25         | Do early exploratory demolition and use structural scans to better understand the structure. Stay flexible in the design so adjustments can be made if conditions differ from drawings. Include a schedule contingency for redesign. |
| 2       | Asbestos and lead abatement discovered during renovation.                                                                                          | 4          | 4      | 16         | Early destructive investigation, confirm scopes with environmental consultants, contingency allowances.                                                                                                                              |
| 3       | Oversaturated market & limited subcontractor bids                                                                                                  | 4          | 4      | 16         | Prequalify bidders early, expand bid pool geographically, early release packages.                                                                                                                                                    |
| 4       | Underpinning Cobb Hall - adjacent historic building stability risk due to potential issues with secant pile wall during excavation                 | 3          | 5      | 15         | Excavate in stages and monitor movement as work progresses. Design conservatively and have backup support options ready if anything starts to shift more than expected.                                                              |
| 5       | Conflicts between tunnel and tie-ins to existing utilities + relocation of high-voltage ductbanks, chilled water, sewer, etc                       | 5          | 4      | 20         | Use 3D coordination and potholing to confirm exact utility locations. Conduct Utility coordination meetings before excavation begins, potholing/locates before excavation, create a phased utility relocation plan.                  |
| 6       | Tight site logistics: no laydown room, limited space for deliveries, and equipment staging leading to schedule delays                              | 5          | 4      | 20         | Use just-in-time deliveries scheduled during off-peak times and clearly defined access points to reduce congestion. Carefully plan material staging and keep site circulation organized.                                             |
| 7       | Crane operations interfering with site access                                                                                                      | 3          | 4      | 12         | Plan crane picks ahead of time and restrict access under the swing radius during lifts. Schedule major lifts during less busy times to reduce disruptions.                                                                           |
| 8       | Conflicts between MEP systems above ceilings due to limited as-built information                                                                   | 4          | 4      | 16         | Use BIM coordination and clash detection before installation begins. Hold coordination meetings with trades to make sure routing works before construction.                                                                          |
| 9       | Concrete work taking longer than estimated                                                                                                         | 3          | 4      | 12         | Track labor productivity during construction and adjust crews if needed. Compare field performance to estimates and update plans if production is slower than expected.                                                              |
| 10      | Material delays due to supply chain issues including long lead items/escalation risk (lighting, specialty finishes, mechanical equipment)          | 4          | 4      | 16         | Order long-lead materials early and identify backup suppliers. Stay flexible with scheduling to account for possible delivery delays. Track via long-lead log, early buyout, secure steel/equipment packages early.                  |
| 11      | Quality issues with detailed architectural finishes including custom millwork, radiused architecture, glass guardrails, terrazzo, acoustic plaster | 3          | 4      | 12         | Create mockups for key finishes and inspect work regularly. Create QA/QC plan to maintain strong quality control throughout construction and use experienced trades.                                                                 |
| 12      | Safety risks to students, staff, and workers                                                                                                       | 5          | 5      | 25         | Use fencing, signage, and clear pedestrian detours to separate people from construction. Enforce safety rules on site and provide regular training.                                                                                  |
| 13      | Schedule delays affecting Fall 2025 Opening deadline requirement to open Cobb & Shumway Halls                                                      | 4          | 5      | 20         | Build a detailed schedule with float for the critical path, do pull-planning, add contingency days. Track progress closely and adjust sequencing or resources if delays start to occur.                                              |

Figure 17. Project Risk Register

## CONCLUSION & DISCUSSION

The McIntire School Expansion project demonstrates the successful delivery of a highly complex renovation and new construction effort within an active and constrained campus environment. The integration of Cobb Hall, a historic structure requiring underpinning and renovation, with the construction of Shumway Hall and the tunnel connection to Rouss-Robertson required careful coordination of structural systems, utilities, and construction sequencing.

Key challenges, including limited site access, dense utility corridors, and the need to maintain ongoing campus and hospital operations, were addressed through detailed planning and phased execution. The selected support-of-excavation system, coordinated tunnel sequencing, and comprehensive site logistics strategy allowed construction activities to proceed safely and efficiently while minimizing disruption to surrounding facilities. Additionally, early identification of risks and implementation of mitigation strategies supported the project's ability to maintain schedule, control costs, and uphold quality standards.

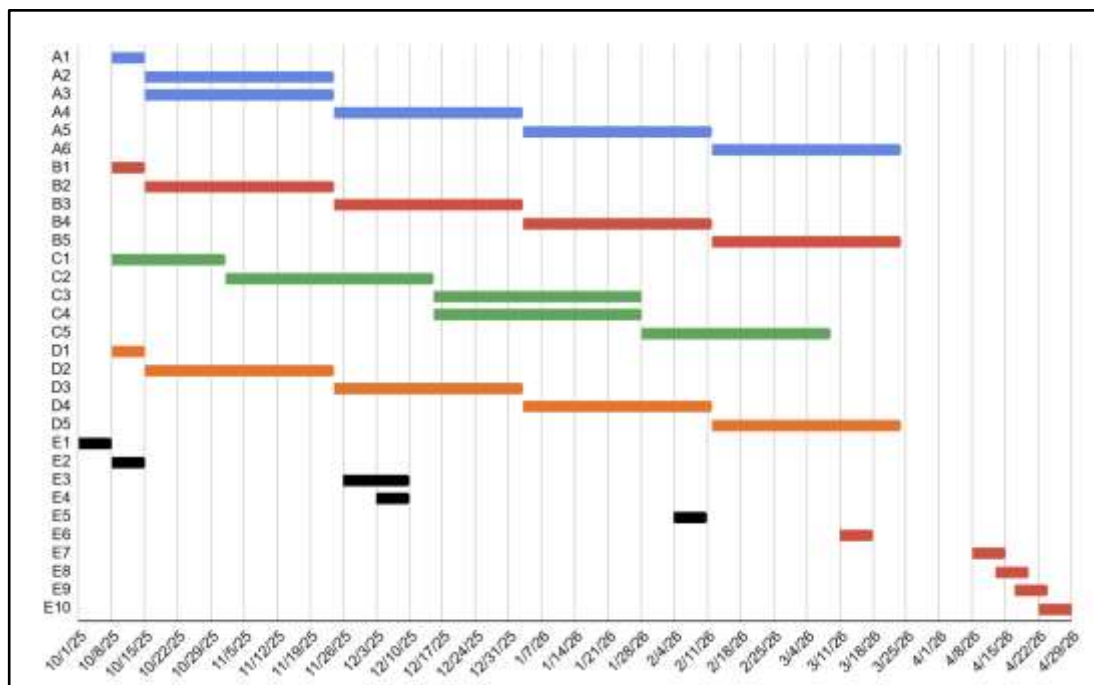
Through strong coordination between design, construction, and planning efforts, the project successfully balanced technical requirements with real-world constructability constraints. The result is a functional and well-integrated expansion that meets the needs of the McIntire School while maintaining the architectural character and operational continuity of the University of Virginia campus

# APPENDICES

## Appendix A - Detailed Schedule

| ID  | Activities                                              | Lead      | Start    | Duration | End      | Dependencies |
|-----|---------------------------------------------------------|-----------|----------|----------|----------|--------------|
| A1  | Identify potential underpinning options                 | Griffin   | 10/8/25  | 7        | 10/15/25 | E1           |
| A2  | Calculate supported load                                | Tim       | 10/15/25 | 40       | 11/24/25 | A1           |
| A3  | Analyze logistics and schedule for underpinning options | Saul      | 10/15/25 | 40       | 11/24/25 | A1           |
| A4  | Create final drawing of chosen support option           | Tim       | 11/24/25 | 40       | 1/3/26   | A1           |
| A5  | Analyze costs for underpinning options                  | Stephanie | 1/3/26   | 40       | 2/12/26  | A1           |
| A6  | Consolidate work and add to risk register               | Griffin   | 2/12/26  | 40       | 3/24/26  | A2-A5        |
| B1  | Review proposed tunnel and analyze risks                | Stephanie | 10/8/25  | 7        | 10/15/25 | E1           |
| B2  | Develop sequencing plan based on chosen method          | Griffin   | 10/15/25 | 40       | 11/24/25 | B1           |
| B3  | Create 3D model of tunnel                               | Saul      | 11/24/25 | 40       | 1/3/26   | B1           |
| B4  | Research and compare tunnel construction methodologies  | Tim       | 1/3/26   | 40       | 2/12/26  | B1           |
| B5  | Consolidate work and add to risk register               | Stephanie | 2/12/26  | 40       | 3/24/26  | B2-B4        |
| C1  | Gather information and determine phasing/staging needs  | Tim       | 10/8/25  | 24       | 11/1/25  | E1           |
| C2  | Develop map for access points and parking               | Stephanie | 11/1/25  | 44       | 12/15/25 | C1           |
| C3  | Analyze and create safety/traffic plan for vehicles     | Griffin   | 12/15/25 | 44       | 1/28/26  | C1           |
| C4  | Analyze and create safety/traffic plan for pedestrians  | Saul      | 12/15/25 | 44       | 1/28/26  | C1           |
| C5  | Consolidate work and add to risk register               | Tim       | 1/28/26  | 40       | 3/8/26   | C2-C4        |
| D1  | Compare self perform vs subcontract approaches          | Saul      | 10/8/25  | 7        | 10/15/25 | E1           |
| D2  | Forecast labor productivity and crew needs              | Tim       | 10/15/25 | 40       | 11/24/25 | D1           |
| D3  | Gather material pricing information                     | Stephanie | 11/24/25 | 40       | 1/3/26   | D1           |
| D4  | Perform quantity take off for concrete                  | Griffin   | 1/3/26   | 40       | 2/12/26  | D1           |
| D5  | Consolidate work and add to risk register               | Saul      | 2/12/26  | 40       | 3/24/26  | D2-D4        |
| E1  | Scheduling assignment                                   | All       | 10/1/25  | 7        | 10/8/25  | N/A          |
| E2  | Interim progress report                                 | All       | 10/8/25  | 7        | 10/15/25 | E1           |
| E3  | End of semester report                                  | All       | 11/26/25 | 14       | 12/10/25 | A1,B1,C1,D1  |
| E4  | End of semester peer review                             | All       | 12/3/25  | 7        | 12/10/25 | A1,B1,C1,D2  |
| E5  | Design Report – Draft #1 (+Constraints)                 | All       | 2/4/26   | 7        | 2/11/26  | E4           |
| E6  | Design Report – Draft #2                                | All       | 3/11/26  | 7        | 3/18/26  | E5           |
| E7  | Poster Submission                                       | All       | 4/8/26   | 7        | 4/15/26  | E6           |
| E8  | Client Presentation                                     | All       | 4/13/26  | 7        | 4/20/26  | E7           |
| E9  | Poster Symposium (in person)                            | All       | 4/17/26  | 7        | 4/24/26  | E8           |
| E10 | Final Design Report                                     | All       | 4/22/26  | 7        | 4/29/26  | E9           |

Appendix A.1 - Original Schedule



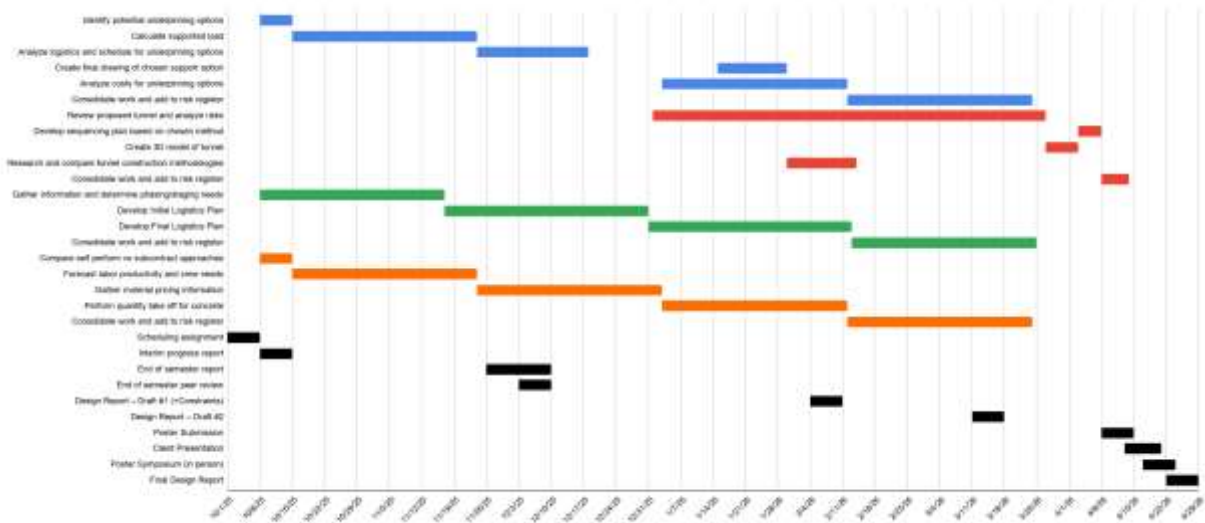
Appendix A.2 - Original Schedule-Driven Gantt Chart

| ID | Activities                                              | Lead             | Start    | Duration | End      | Dependencies          |
|----|---------------------------------------------------------|------------------|----------|----------|----------|-----------------------|
| A1 | Identify potential underpinning options                 | Griffin          | 10/8/25  | 7        | 10/15/25 | E1                    |
| A2 | Calculate supported load                                | Tim              | 10/15/25 | 40       | 11/24/25 | A1                    |
| A3 | Analyze logistics and schedule for underpinning options | Tim              | 11/24/25 | 24       | 12/18/25 | A1                    |
| A4 | Create final drawing of chosen support option           | Tim              | 1/15/26  | 15       | 1/30/26  | A1                    |
| A5 | Analyze costs for underpinning options                  | Aden             | 1/3/26   | 40       | 2/12/26  | A1                    |
| A6 | Consolidate work and add to risk register               | Griffin          | 2/12/26  | 40       | 3/24/26  | A2-A5                 |
| B1 | Review proposed tunnel and analyze risks                | Aden             | 1/1/26   | 85       | 3/27/26  | E1                    |
| B2 | Develop sequencing plan based on method                 | Aden             | 4/3/26   | 5        | 4/8/26   | B3                    |
| B3 | Create 3D model of tunnel                               | Aden             | 3/27/26  | 7        | 4/3/26   | B1, B4                |
| B4 | Research and compare tunnel construction methodologies  | Stephanie        | 1/30/26  | 15       | 2/14/26  | E1                    |
| B5 | Consolidate work and add to risk register               | Stephanie / Aden | 4/8/26   | 6        | 4/14/26  | B2                    |
| C1 | Gather information and determine phasing/staging needs  | Tim              | 10/8/25  | 40       | 11/17/25 | E1                    |
| C2 | Develop Initial Logistics Plan                          | All              | 11/17/25 | 44       | 12/31/25 | C1                    |
| C3 | Develop Final Logistics Plan                            | Stephanie        | 12/31/25 | 44       | 2/13/26  | C1                    |
| C4 | Consolidate work and add to risk register               | Stephanie        | 2/13/26  | 40       | 3/25/26  | C2-C3                 |
| D1 | Compare self perform vs subcontract approaches          | Saul             | 10/8/25  | 7        | 10/15/25 | E1                    |
| D2 | Forecast labor productivity and crew needs              | Griffin          | 10/15/25 | 40       | 11/24/25 | D1                    |
| D3 | Gather material pricing information                     | Griffin          | 11/24/25 | 40       | 1/3/26   | D1                    |
| D4 | Perform quantity take off for concrete                  | Griffin          | 1/3/26   | 40       | 2/12/26  | D1                    |
| D5 | Consolidate work and add to risk register               | Saul             | 2/12/26  | 40       | 3/24/26  | D2-D4                 |
| E1 | Scheduling assignment                                   | All              | 10/1/25  | 7        | 10/8/25  | N/A                   |
| E2 | Interim progress report                                 | All              | 10/8/25  | 7        | 10/15/25 | E1                    |
| E3 | End of semester report                                  | All              | 11/26/25 | 14       | 12/10/25 | A,B,C,D1              |
| E4 | End of semester peer review                             | All              | 12/3/25  | 7        | 12/10/25 | A,B1;C,D <sub>2</sub> |
| E5 | Design Report – Draft #1 (+Constraints)                 | All              | 2/4/26   | 7        | 2/11/26  | E4                    |
| E6 | Design Report – Draft #2                                | All              | 3/11/26  | 7        | 3/18/26  | E5                    |
| E7 | Poster Submission                                       | All              | 4/8/26   | 7        | 4/15/26  | E6                    |



|            |                              |     |         |   |         |    |
|------------|------------------------------|-----|---------|---|---------|----|
| <b>E8</b>  | Client Presentation          | All | 4/13/26 | 8 | 4/21/26 | E7 |
| <b>E9</b>  | Poster Symposium (in person) | All | 4/17/26 | 7 | 4/24/26 | E8 |
| <b>E10</b> | Final Design Report          | All | 4/22/26 | 7 | 4/29/26 | E9 |

**Appendix A.3 - Updated Schedule**



**Appendix A.4 - Updated Schedule-Driven Gantt Chart**

## Appendix B - Design Evolution



### Appendix B.1 - Original version of the site work phasing plan

In this initial conceptual iteration, the designated traffic route was determined to be too narrow to accommodate the required construction equipment and haul vehicles safely. This posed potential safety and operational challenges, including limited maneuverability, increased risk of conflicts between vehicles, and restricted access for efficient material transport. As a result, the plan was revised to provide wider, clearly defined routes that better support safe and efficient site operations within the constrained campus environment.



## **Appendix C - Engineering Standards Compliance**

Throughout the McIntire School Expansion project, applicable engineering standards were directly incorporated into the team's analysis, design decisions, and construction planning to ensure compliance with code requirements and realistic constructability. Rather than being referenced generically, these standards were used to guide specific components of the project, including underpinning design, tunnel sequencing, utility coordination, and concrete estimating.

### **Virginia Uniform Statewide Building Code (VUSBC)**

The Virginia Uniform Statewide Building Code was used as the baseline for ensuring compliance with local building requirements. This was most directly reflected in the planning of excavation and support-of-excavation systems, where protection of the existing Cobb Hall structure was a critical requirement. Code considerations also informed the approach to temporary support, structural safety during phased construction, and maintaining safe conditions within an active campus environment.

### **American Concrete Institute (ACI 318)**

ACI 318 requirements were applied in the development of the cast-in-place concrete estimate and quantity takeoffs. Assumptions related to reinforcing bar spacing, concrete placement, and durability were based on ACI guidelines, particularly for elements in contact with soil such as footings and slabs. These requirements helped ensure that the estimated quantities and constructability assumptions aligned with realistic construction practices.

### **American Institute of Steel Construction (AISC)**

AISC standards were referenced in evaluating temporary support systems and potential steel components associated with excavation support and tunnel construction. While detailed steel design was not the primary focus of this project, AISC guidance informed decisions related to stability, constructability, and coordination between structural systems.

### **ASCE and FHWA Support-of-Excavation (SOE) Guidance**

ASCE and FHWA guidance played a central role in the selection and evaluation of the support-of-excavation system for Cobb Hall. These standards informed the comparison between secant pile walls, soil nailing, and slurry walls, particularly in terms of constructability, groundwater control, and performance near existing structures. Criteria such as wall stiffness, deformation control, and compatibility with a constrained site directly influenced the selection of the secant pile wall as the preferred system.

### **NFPA 70 (National Electrical Code) and NFPA 90A**

NFPA standards were applied during the planning of tunnel construction and utility coordination. These codes informed required clearances and safety considerations when routing electrical and mechanical systems through a constrained underground corridor. This was especially important when evaluating how new utilities could be installed above the tunnel while maintaining safe separation from existing systems.

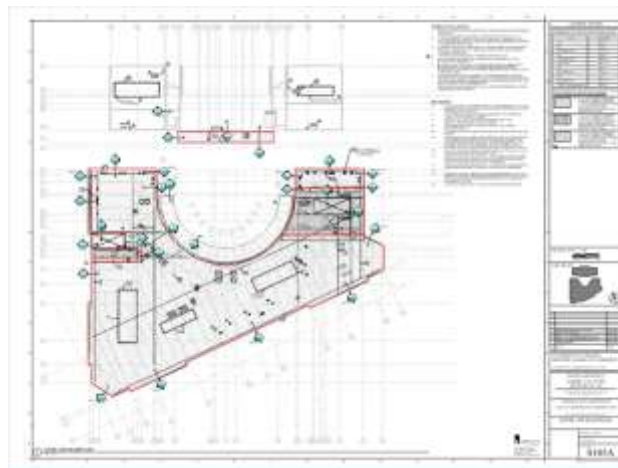
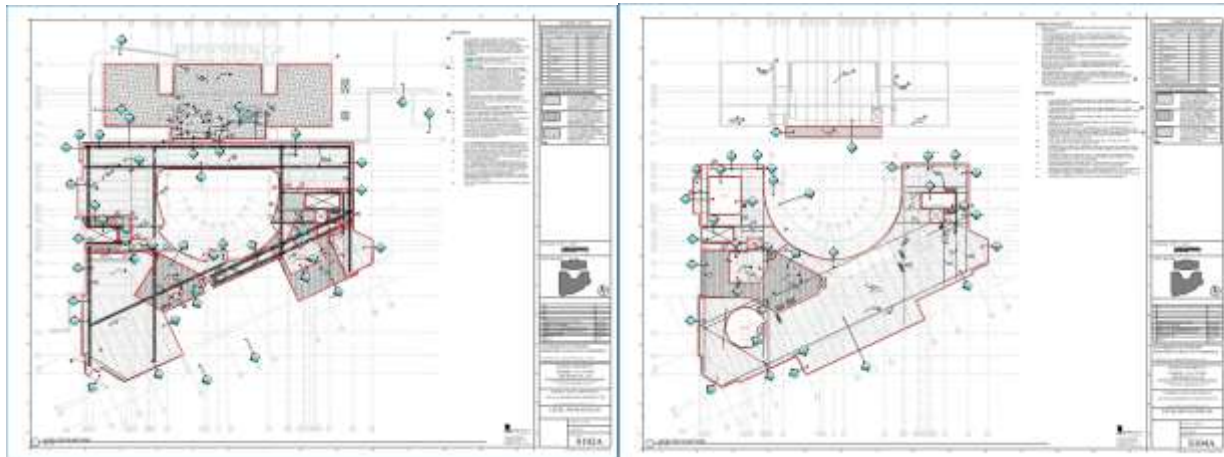
### **ASCE 38-22 (Standard for Utility Investigations)**

ASCE 38-22 was directly applied in the utility coordination and clash detection process. The standard guided the identification and verification of existing utilities and supported the need for potholing and field verification prior to excavation. This informed the tunnel sequencing plan and ensured that utility conflicts could be addressed through relocation or protection strategies before construction progressed.

### **ASCE 7-22 (Minimum Design Loads)**

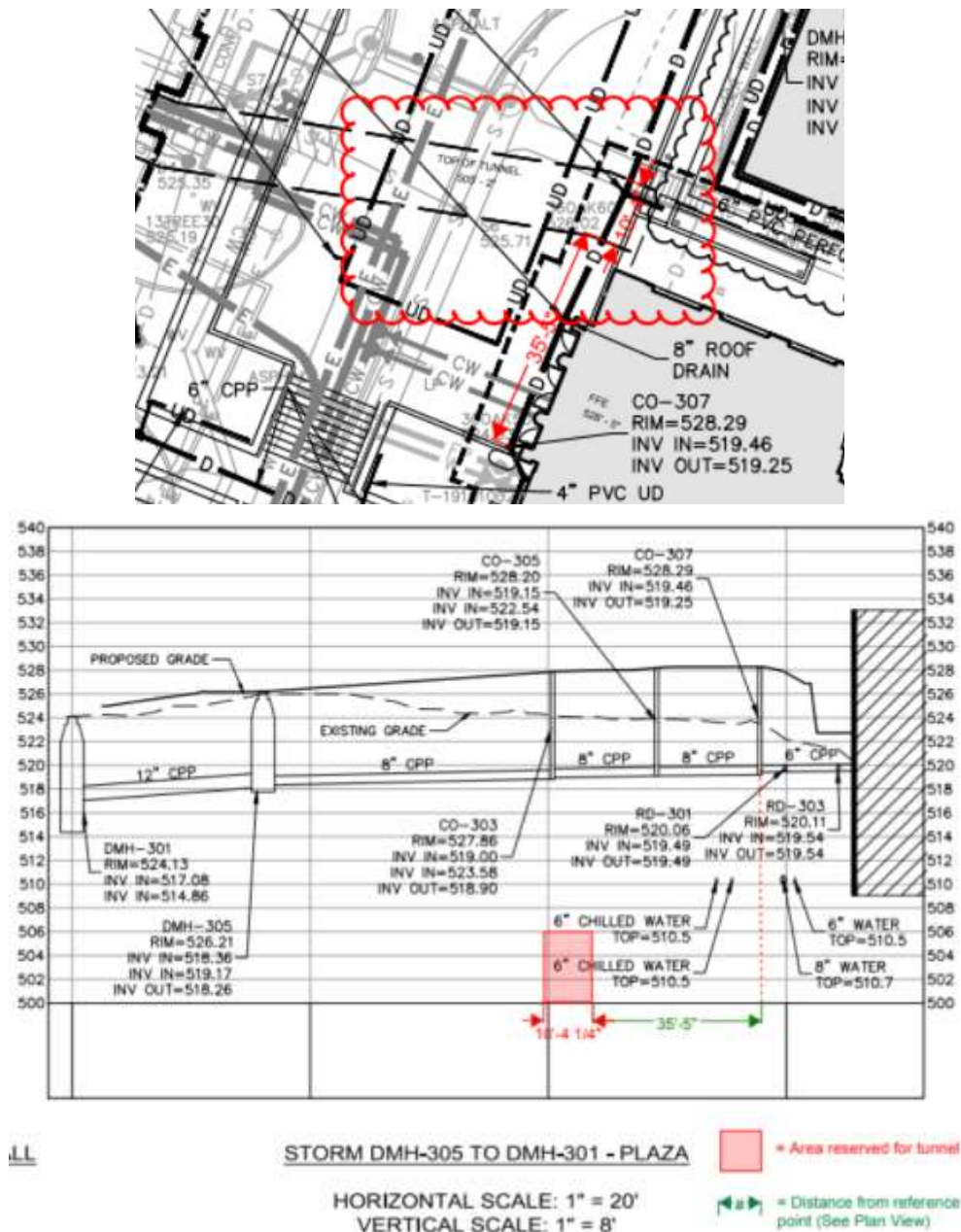
ASCE 7-22 was used in the underpinning analysis to determine governing load conditions. Vertical loads from Cobb Hall and lateral earth pressures were calculated using ASCE load combinations, which informed the design and evaluation of the support-of-excavation system. These calculations ensured that the selected system could safely support existing structures during excavation.

## Appendix D - Supporting Technical Deliverables



**Appendix D.1 - Screenshots from the concrete quantity survey**

## Appendix E - Clash Detection Details (New Utilities)



Appendix E.1 - New Storm Drainage Verification (Left: Plan View; Right: Profile View)



44



**ELECTRIC DUCT BANK UNDER RUPPEL DRIVE PROFILE VIEW**

**Appendix E.3 - New Electrical DB Verification (Top: Plan View; Bottom: Profile View)**

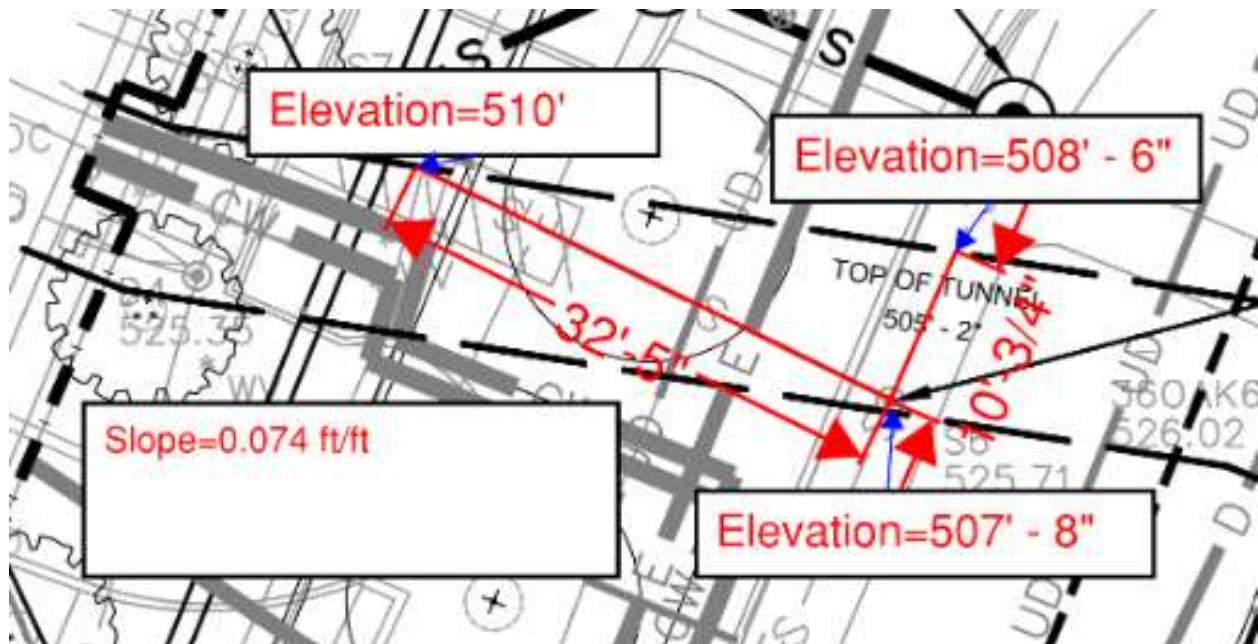
| Utility Type | Sheets | Detail | Notes |
|--------------|--------|--------|-------|
|--------------|--------|--------|-------|

|               |               |                                                     |                                                                                                                                                                                                                                                                                           |
|---------------|---------------|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storm Drain   | C103/C<br>300 | C300: DMH 305 to<br>DMH 301 - Plaza                 | Storm DMH 305 - DMH 301; Storm drainage crosses over tunnel from CO - 305 -> DMH 305; No clashes with tunnel, sits above 506'                                                                                                                                                             |
| Underdrain    | C103/C<br>400 | C400: Underdrain<br>Detail                          | C103 states the elevation at 519.35', and no additional elevations are provided; Assuming constant elevation, no clashes are detected. Additionally, storm drains require a maximum 3 ft depth from the finished grade, which falls above inv. out elevation; assume the lowest elevation |
| Chilled Water | C104/C<br>304 | C304: Chilled Water -<br>Ripple Drive               | Tunnel sits below the potential clash section; No clashes with tunnel                                                                                                                                                                                                                     |
| Electrical    | C105/C<br>302 | C302: Electrical Duct<br>Bank Under Ripple<br>Drive | Tunnel site between where the final grade is 527'->528'; whole electrical section sits above 506'                                                                                                                                                                                         |
| Water         | C102          | N/A                                                 | Water lines do not constrain tunnel construction                                                                                                                                                                                                                                          |
| Sewer         | C102          | N/A                                                 | Sewer lines do not constrain tunnel construction                                                                                                                                                                                                                                          |

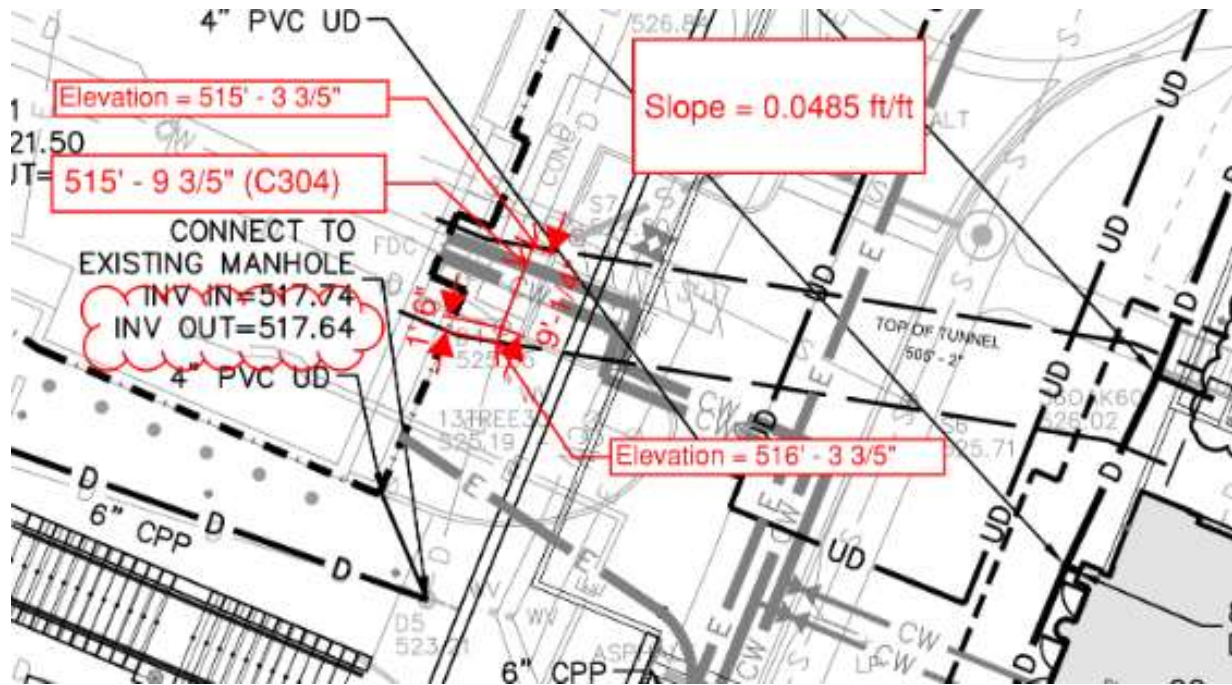
#### Appendix E.4 - New Utilities Notes



## Appendix F - Clash Detection Details (Existing Utilities)



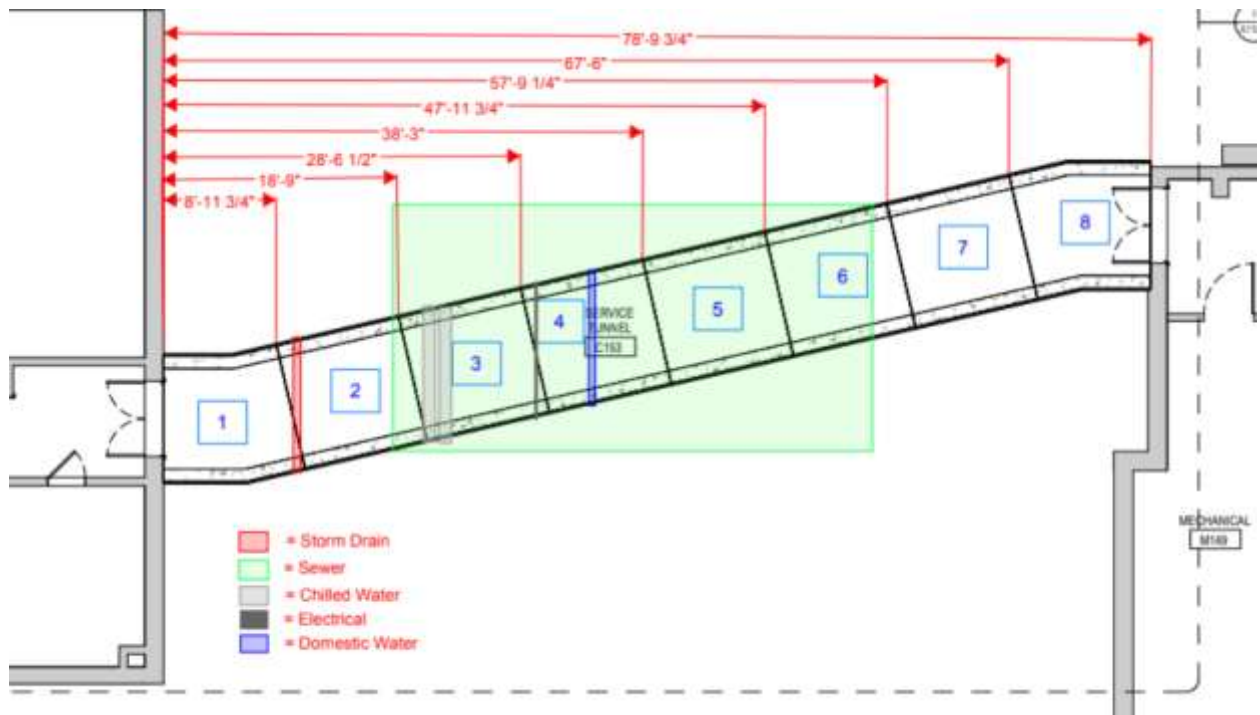
Appendix F.1 - Existing Sewage Line (Plan View with Elevations)



| Utility Type                                 | Sheets          | Details                                | Notes                                                                                                  |
|----------------------------------------------|-----------------|----------------------------------------|--------------------------------------------------------------------------------------------------------|
| Storm Drain                                  | C200A/C304/C103 | C304: Chilled Water Rurple Drive       | Assume constant slope 0.0485 ft/ft; Storm drain sits at 515 - 9 3/5" at lowest - no clash              |
| Chilled Water                                | C200A           | N/A                                    | Assume pressurized/constant elevation; Chilled Water connects with existing tunnel at ~521' - no clash |
| Electrical                                   | C200A/C302      | C302: Electrical Ductbank Under Rurple | Assume constant elevation; Installed ductbank connects to existing at 520 ft - no clash                |
| Water                                        | C200A/C304      | C304: Hydrant - Plaza                  | Assume pressurized (constant elevation) - Water line sits at 522' - no clash                           |
| Sewer                                        | C200A/C102/C304 | C304: Sewer Plaza                      | Assume constant slope 0.074 ft/ft; Sewer sits at 507' - 8" lowest - no clash                           |
| <i>*No utility shall be higher than 527'</i> |                 |                                        |                                                                                                        |

### **Appendix F.3 - Existing Utilities Notes**

## Appendix G - Existing Utility/Tunnel Sequencing Supporting Graphics



**Appendix G.1 - Plan View of Existing Utilities Versus Tunnel Location**

## Appendix H - Underpinning Support Design Recommendations

| Boring No. | Sample Depth (Feet) | Natural Water Content (%) | Liquid Limit/ Plasticity Index | % Passing No. 200 Sieve | USCS Class. |
|------------|---------------------|---------------------------|--------------------------------|-------------------------|-------------|
| B-01       | 8.5-10              | 16.6                      | --                             | --                      | --          |
| B-01       | 18.5-20             | 22.8                      | --                             | --                      | --          |
| B-01       | 33.5-35             | 32.7                      | NP/NP                          | 66.6                    | ML          |
| B-01       | 48.5-50             | 36.3                      | --                             | --                      | --          |
| Boring No. | Sample Depth (Feet) | Natural Water Content (%) | Liquid Limit/ Plasticity Index | % Passing No. 200 Sieve | USCS Class. |
| B-03       | 5-6.5               | 24.7                      | 44/5                           | 80.6                    | ML          |
| B-03       | 13.5-15             | 21.0                      | --                             | --                      | --          |
| B-03       | 23.5-25             | 13.4                      | --                             | --                      | --          |
| B-03       | 33.5-35             | 25.9                      | --                             | --                      | --          |
| B-04       | 5-6.5               | 22.6                      | 58/14                          | 80.2                    | MH          |
| B-04       | 13.5-15             | 9.7                       | --                             | --                      | --          |
| B-04       | 28.5-30             | 20.8                      | --                             | --                      | --          |
| B-04       | 43.5-45             | 32.6                      | --                             | --                      | --          |
| B-07       | 8.5-10              | 17.2                      | NP/NP                          | 70.1                    | ML          |
| B-07       | 18.5-20             | 16.7                      | --                             | --                      | --          |
| B-08       | 2.5-4               | 17.9                      | --                             | --                      | --          |
| B-08       | 13.5-15             | 14.9                      | NP/NP                          | 44.7                    | SM          |
| B-08       | 23.5-25             | 19.0                      | --                             | --                      | --          |
| B-09       | 6-8                 | 28.0                      | --                             | 86.4                    | ML          |
| B-10       | 13-15               | 27.3                      | --                             | 76.4                    | ML          |
| B-11       | 23-25               | 21.6                      | --                             | 89.6                    | ML          |

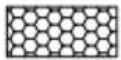
Appendix H.1 - Geotechnical Design Site Soil Classification

| Soil Type                        | Base Friction Coefficient | Lateral Earth Pressure Coefficient (k) |        | Equivalent Fluid Unit Weight ( $\gamma_{eq}$ , pcf) |        |         |
|----------------------------------|---------------------------|----------------------------------------|--------|-----------------------------------------------------|--------|---------|
|                                  |                           | At-rest                                | Active | At-rest                                             | Active | Passive |
| Sandy SILT (ML) or more granular | 0.34                      | 0.53                                   | 0.36   | 64                                                  | 43     | 300     |

The above earth pressure coefficients can also be used for the design of site retaining walls, assuming that sandy SILT (ML) or more granular soils are utilized as backfill.

A moist unit weight of 120 pcf for soil should be used for design calculations. The backfill material should be extended a minimum distance of 0.5 times the wall height laterally from the back face of the wall, or for a cantilevered wall, from the heel of the wall footing.

## Appendix H.2 - Geotechnical Design Recommended Characteristics

| MARK                                                                                | DESCRIPTION                        | SUPERIMPOSED LOADS |     |
|-------------------------------------------------------------------------------------|------------------------------------|--------------------|-----|
|                                                                                     |                                    | SDL                | LL  |
|  | SHUMWAY FLOOR - LEVEL 200 - 400    | 10                 | 100 |
|  | SHUMWAY FLOOR - LEVEL 500          | 10                 | 150 |
|  | SHUMWAY ROOF                       | 45                 | 20  |
|  | COBB OFFICE/ CLASSROOM/ MECHANICAL | 10                 | 65  |
|  | COBB ASSEMBLY                      | 10                 | 100 |
|  | COBB ROOF                          | 25                 | 20  |
|  | SHUMWAY PLANTERS                   | 240                | 100 |
|  | SHUMWAY COURTYARD                  | 120                | 100 |
|  | SHUMWAY SOG AND COBB               | 20                 | 100 |

## Appendix H.3 - Live and Dead Loading Plans

## Appendix I - Supporting Materials

All supplementary materials associated with this capstone project, including documents not embedded within the report, are available at the following link:

[https://drive.google.com/drive/folders/1F353GQSILqkL34P2-5zSIEQPy6yYj0Ln?usp=drive\\_link](https://drive.google.com/drive/folders/1F353GQSILqkL34P2-5zSIEQPy6yYj0Ln?usp=drive_link)



## Appendix J - Peer Review

| Project Member   | Roles & Responsibilities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stephanie Castro | <ul style="list-style-type: none"> <li>● Drafted and formatted report</li> <li>● Project Complexity</li> <li>● Risk register</li> <li>● Logistics Site Plans</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Griffin Harrison | <ul style="list-style-type: none"> <li>● Performed quantity takeoffs for the concrete scope</li> <li>● Developed initial site logistics plan considerations</li> <li>● Performed RSM means cost and labor analysis for the concrete scope including formwork and rebar</li> <li>● Developed underpinning design considerations</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Timothy Cheng    | <ul style="list-style-type: none"> <li>● Drafted underpinning drawings plan views</li> <li>● Drafted underpinning design drawing, typical section</li> <li>● Performed calculations for SOE underpinning requirements</li> <li>● Developed underpinning design considerations</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Saul Malca       | <ul style="list-style-type: none"> <li>● Completed subcontractor scoping requirements</li> <li>● Developed concrete estimate tables and costs</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Aden Melton      | <ul style="list-style-type: none"> <li>● Developed cost and schedule analysis for SOE and underpinning options</li> <li>● Drafted and revised sections: <ul style="list-style-type: none"> <li>○ Support of Excavation Analysis</li> <li>○ Tunnel Design Considerations and Codes</li> <li>○ Tunnel Sequencing Plan</li> </ul> </li> <li>● Verified coordination between new utilities, existing utilities, and tunnel</li> <li>● Drafted Appendix E - New Utility Details</li> <li>● Drafted Appendix F - Existing Utility Details</li> <li>● Drafted Appendix G - Sequence Plan Supporting Drawings</li> <li>● Drafted SOE placement for tunnel construction</li> <li>● Revised appendices and references</li> <li>● Generated Revit model of tunnel with utilities</li> <li>● Created sequencing step table <ul style="list-style-type: none"> <li>○ Identified utilities to be out</li> <li>○ Identified order of tunnel pieces</li> <li>○ Identified potential risks and activities of highest and lowest risk</li> </ul> </li> </ul> |